

06026240 INTELLIGENT SYSTEM DEVELOPMENT

Chapter II

Introduction to Computer Vision

with Dr. Bhattarabhorn Wattanacheep

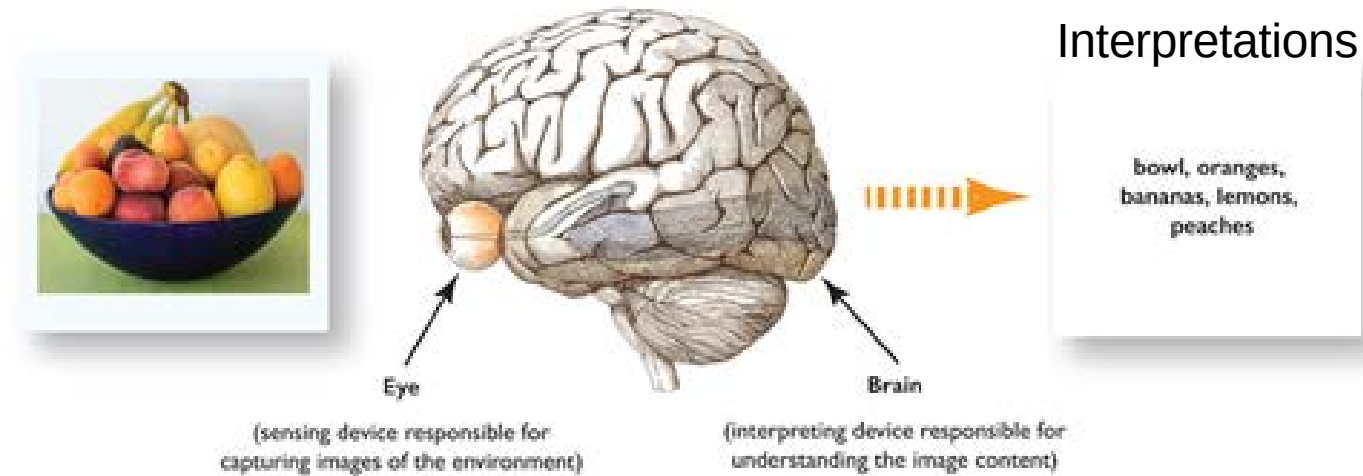
Today's Agenda — Week 2



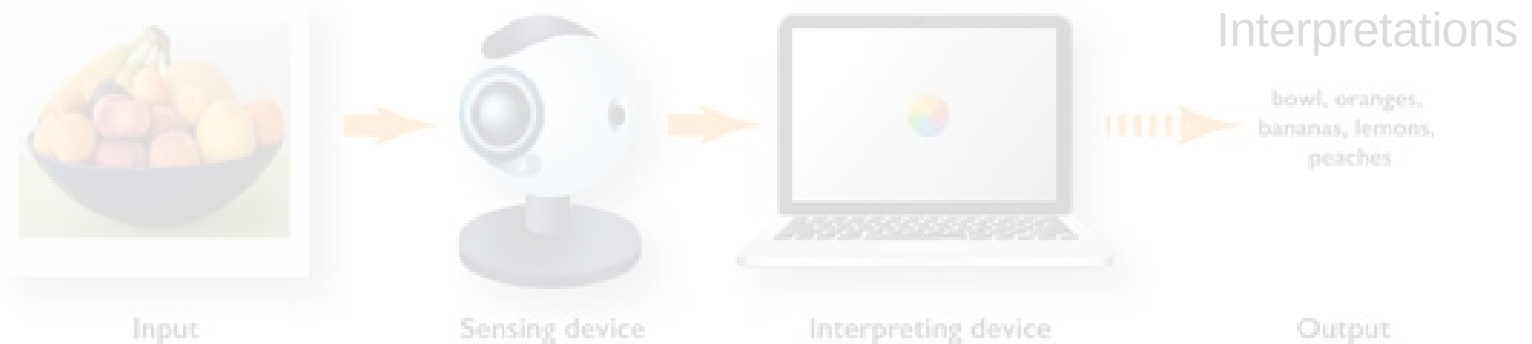
- 01 What is Computer Vision?
- 02 Computer Vision in the Real World
- 03 Image Filtering & Convolution
- 04 Image Transformations
- 05 Camera Matrices → 3D Reconstruction

What is (computer) vision?

Human Vision System



Computer Vision System



<https://www.spiceworks.com/tech/artificial-intelligence/articles/what-is-computer-vision/>

- To bridge the gap between pixels and “meaning”

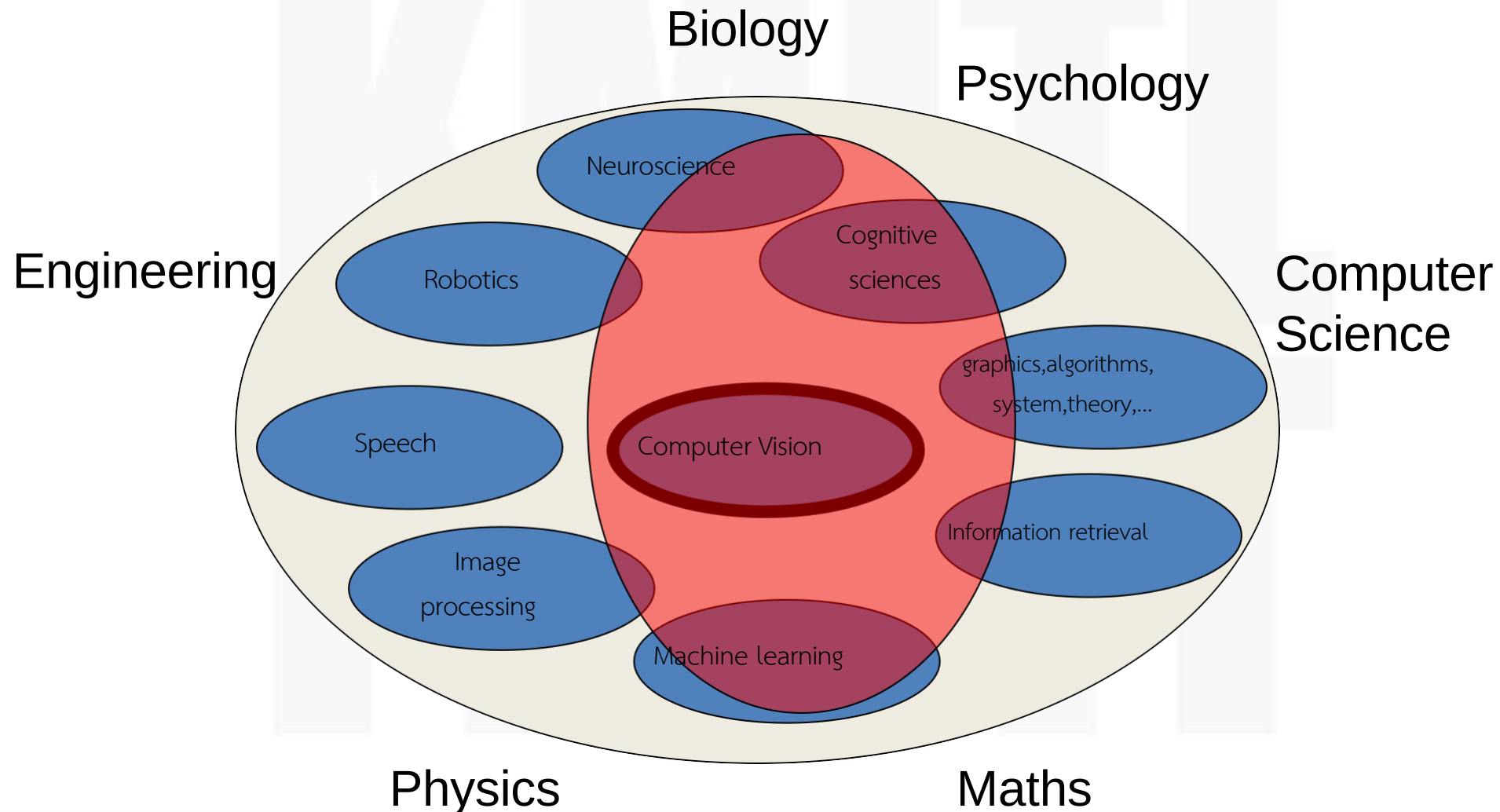


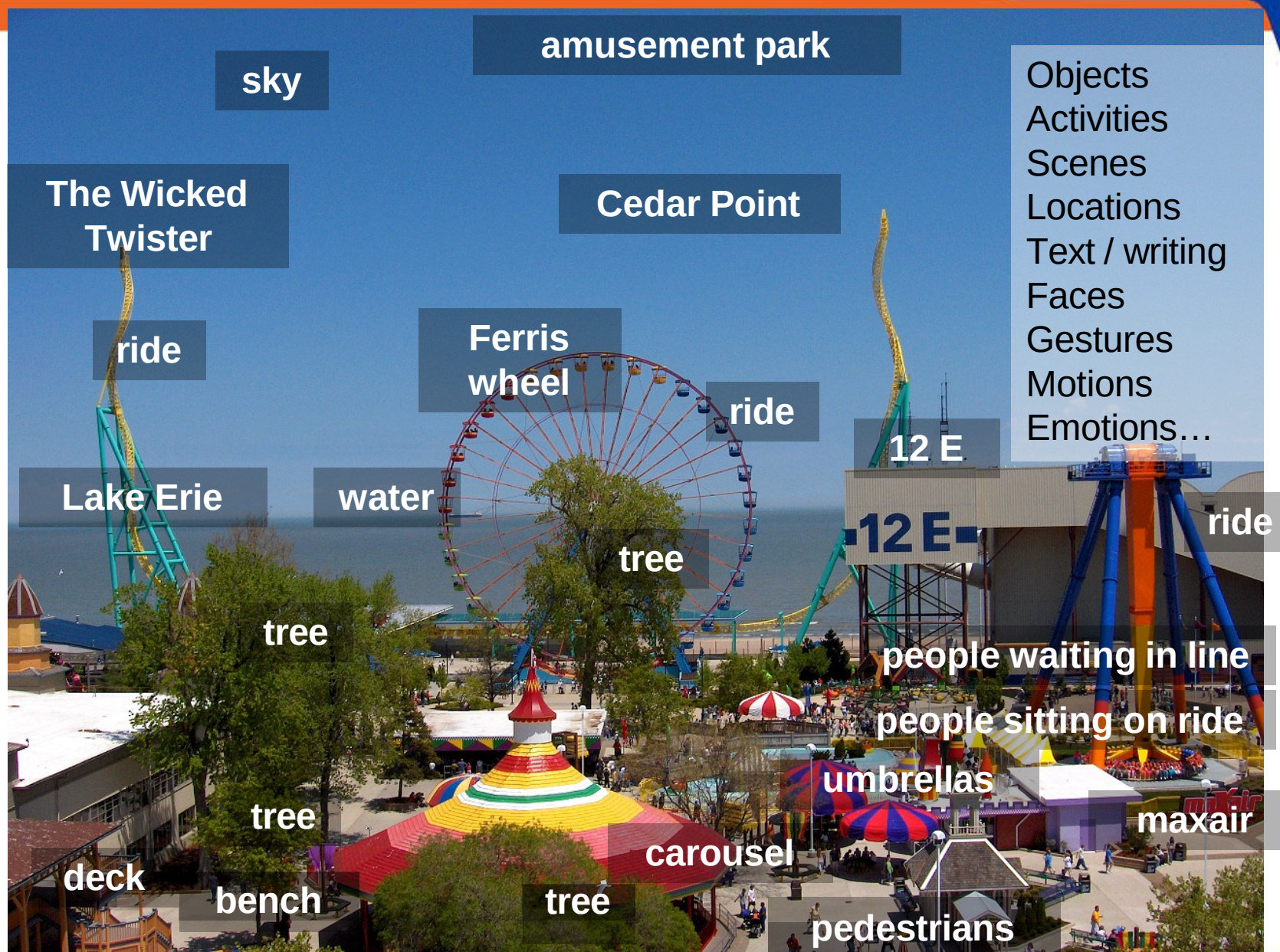
What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What a computer sees

What is it related to?





Objects
Activities
Scenes
Locations
Text / writing
Faces
Gestures
Motions
Emotions...

Computer Vision Tasks

Classification



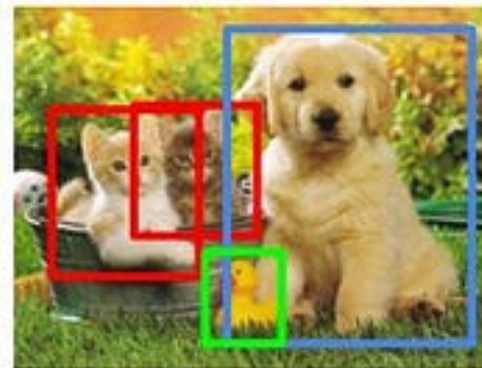
CAT

**Classification
+ Localization**



CAT

Object Detection



CAT, DOG, DUCK

Instance Segmentation

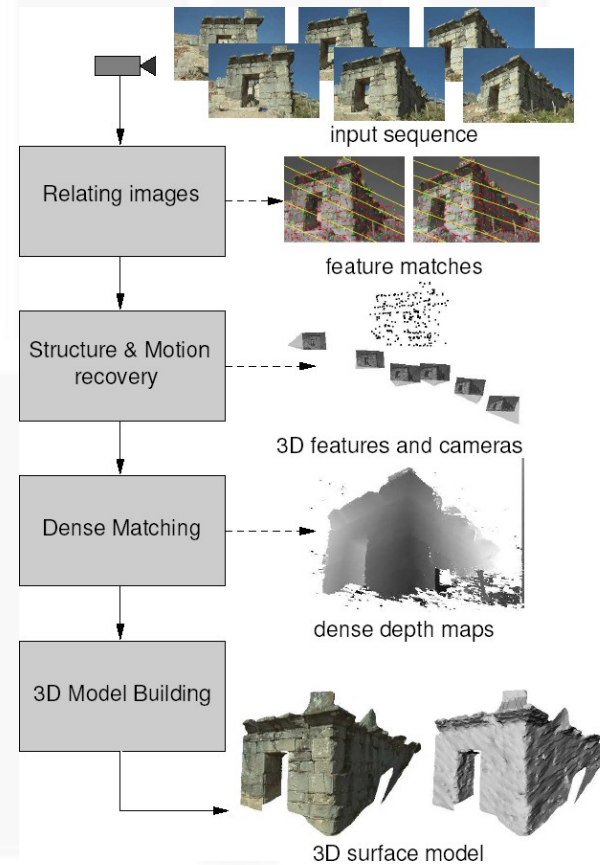


CAT, DOG, DUCK

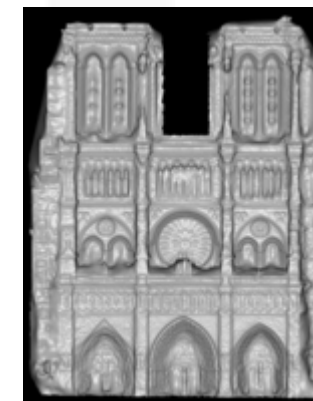
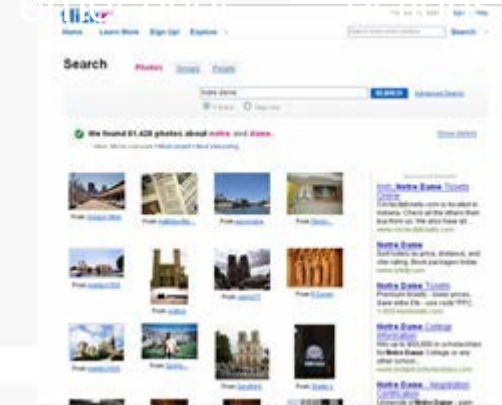
Single object

Multiple objects

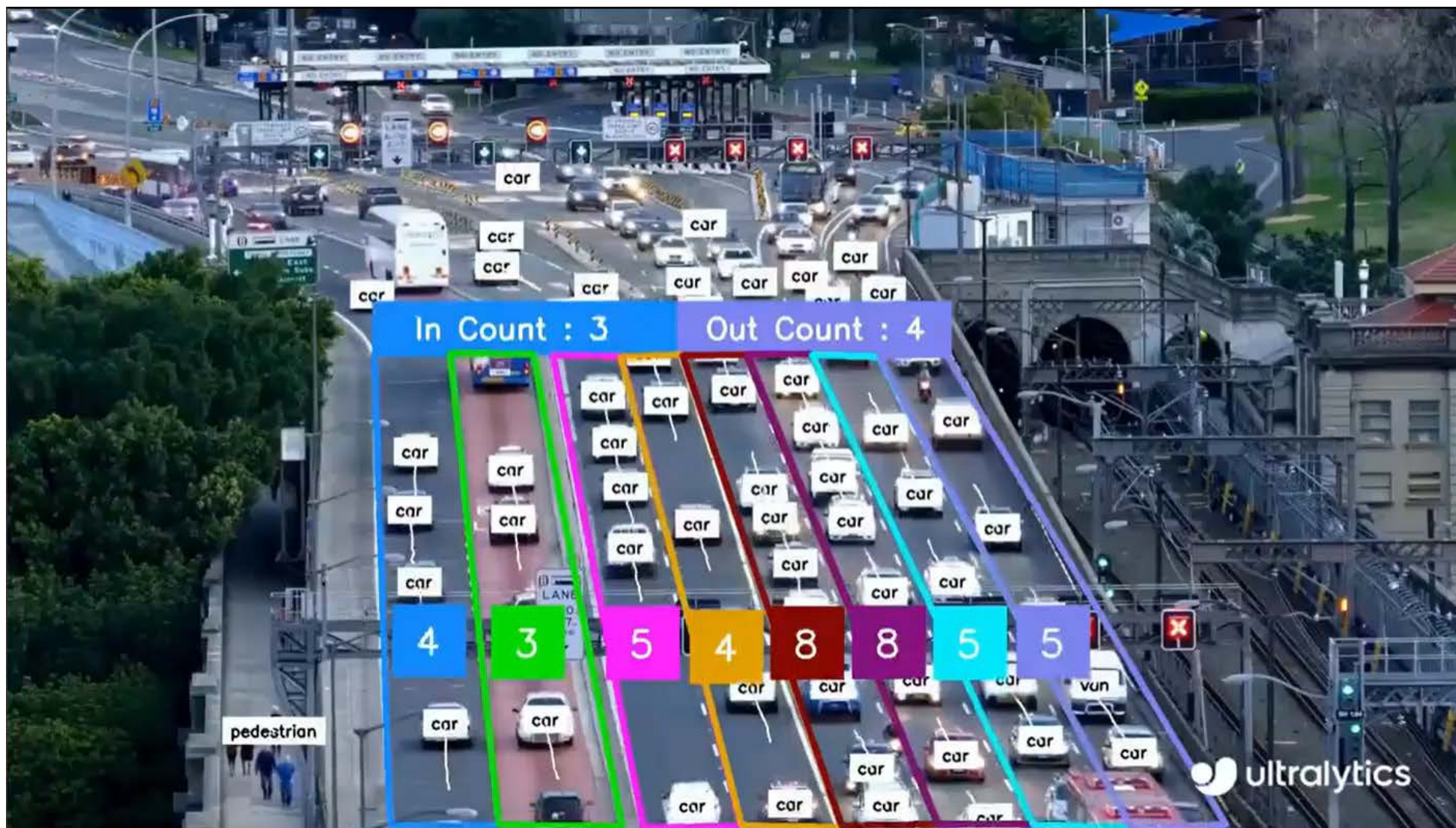
Vision as measurement device

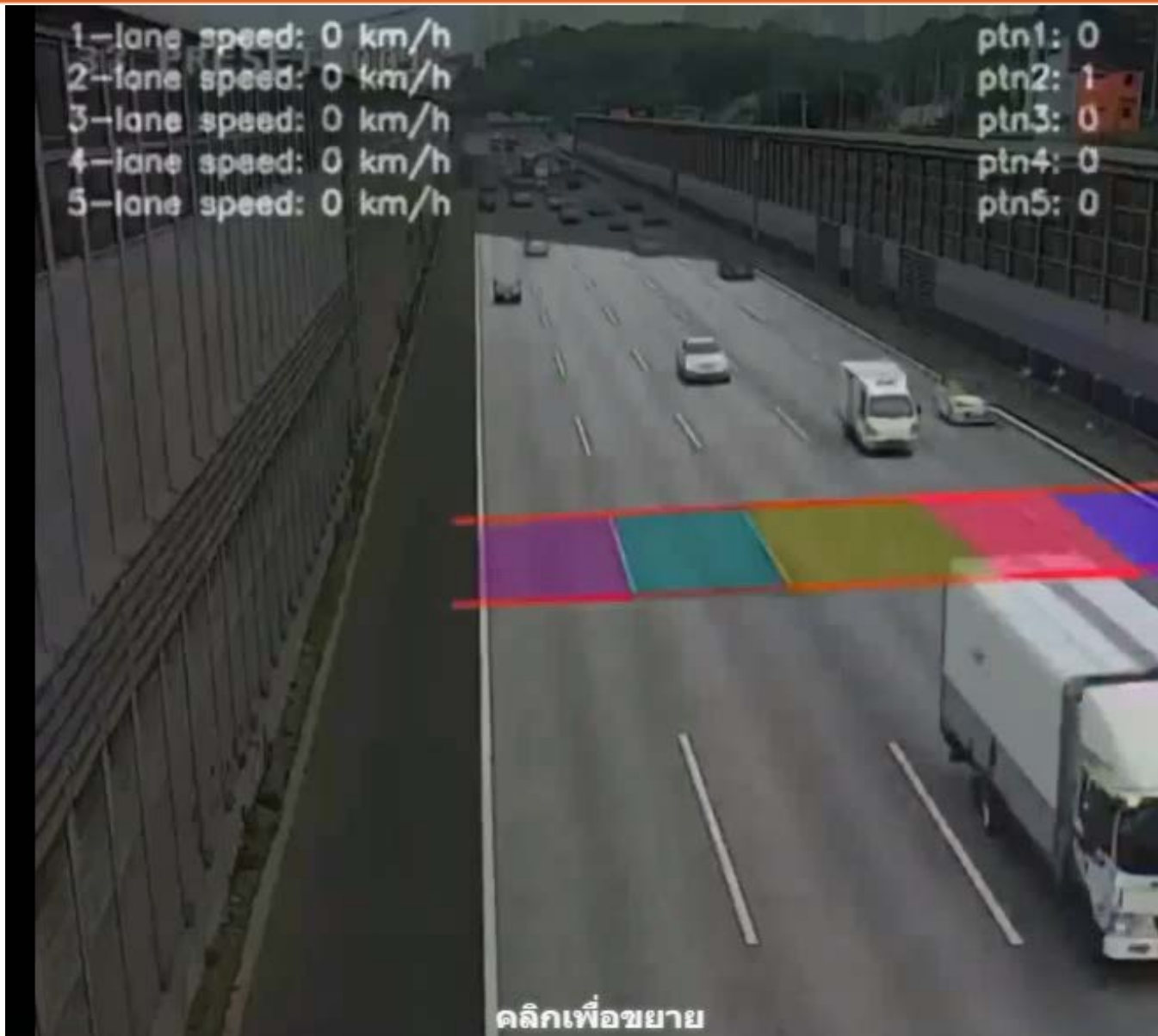


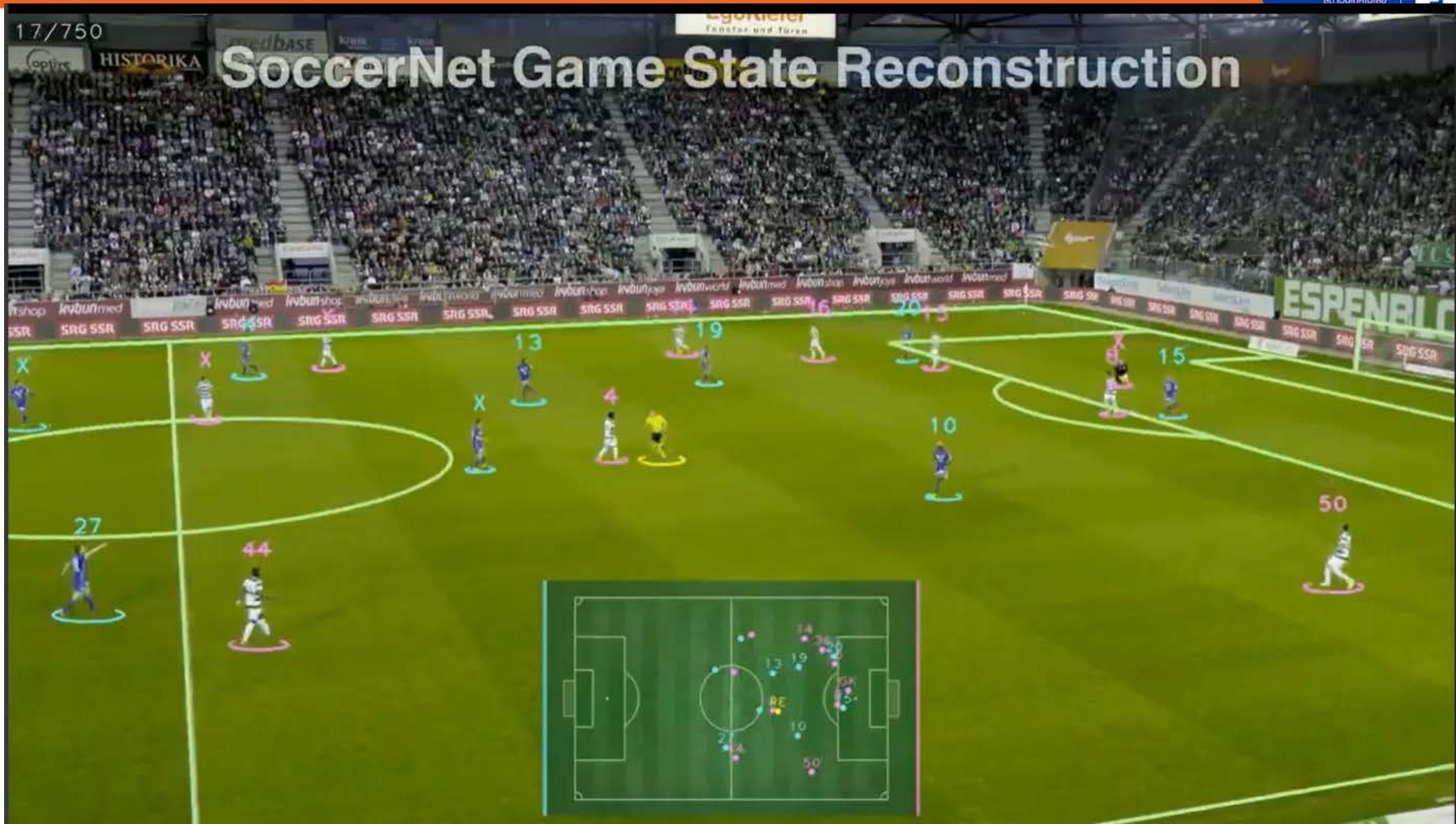
Pollefeys et al.

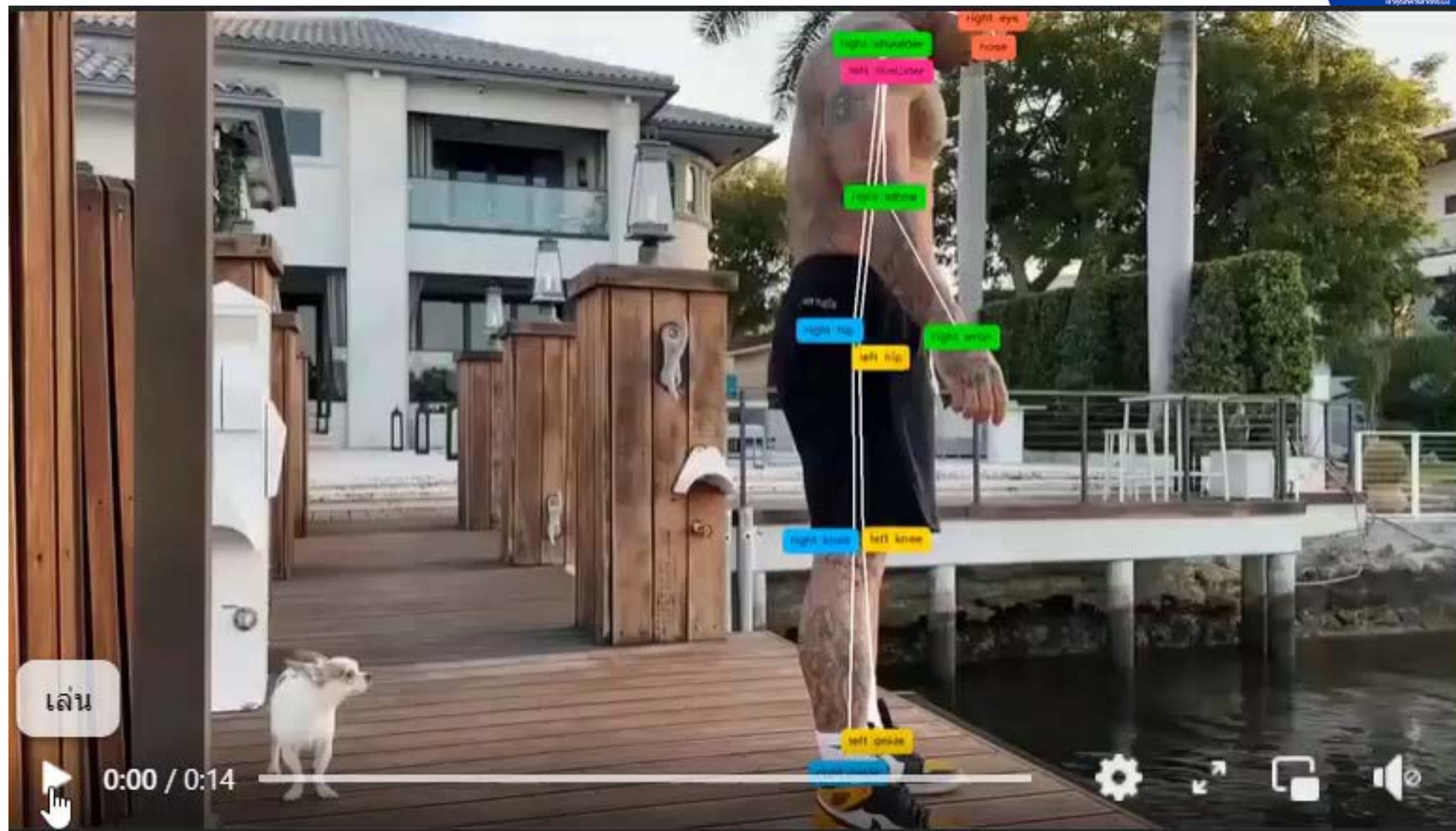


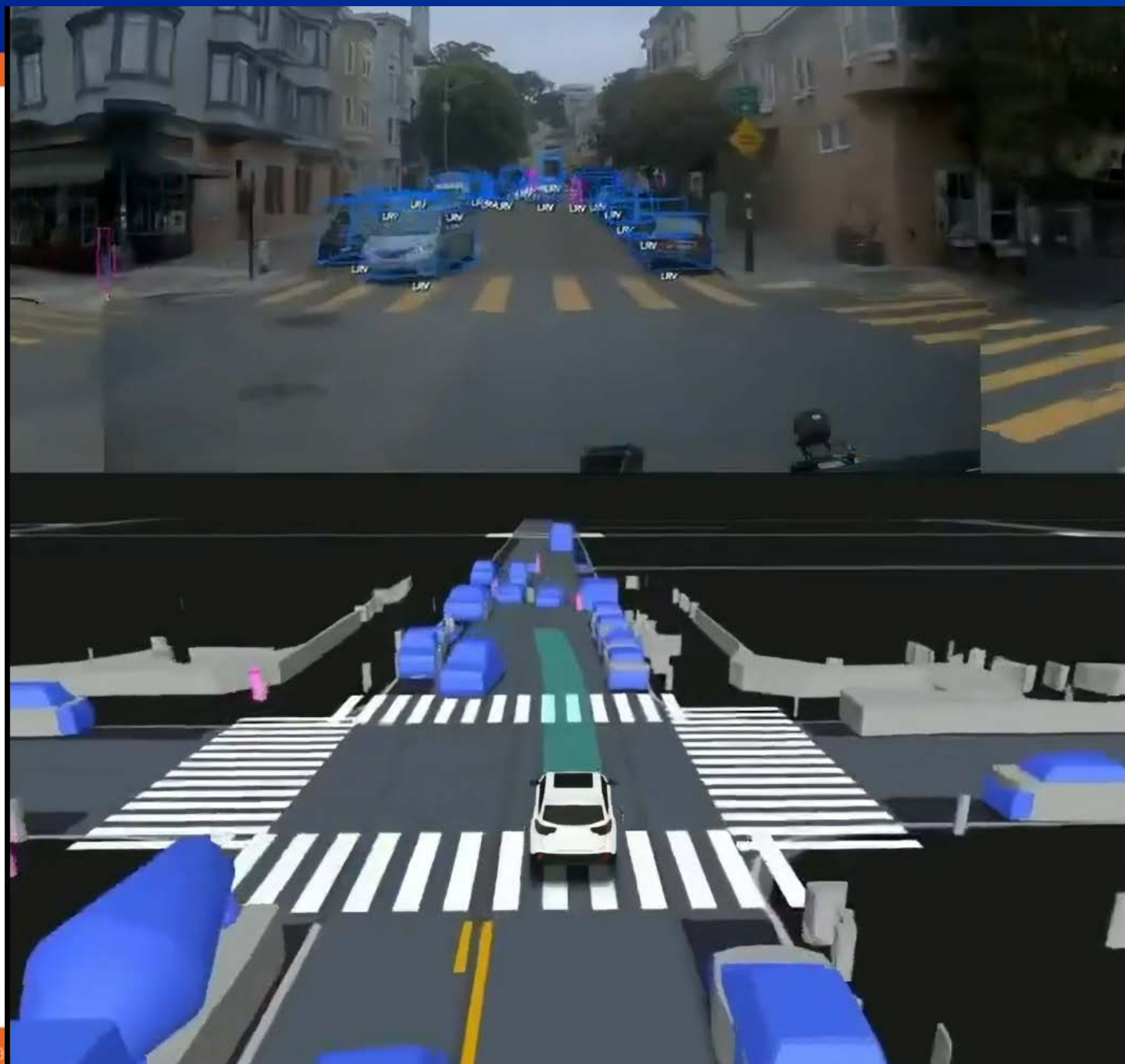
Goesele et al.

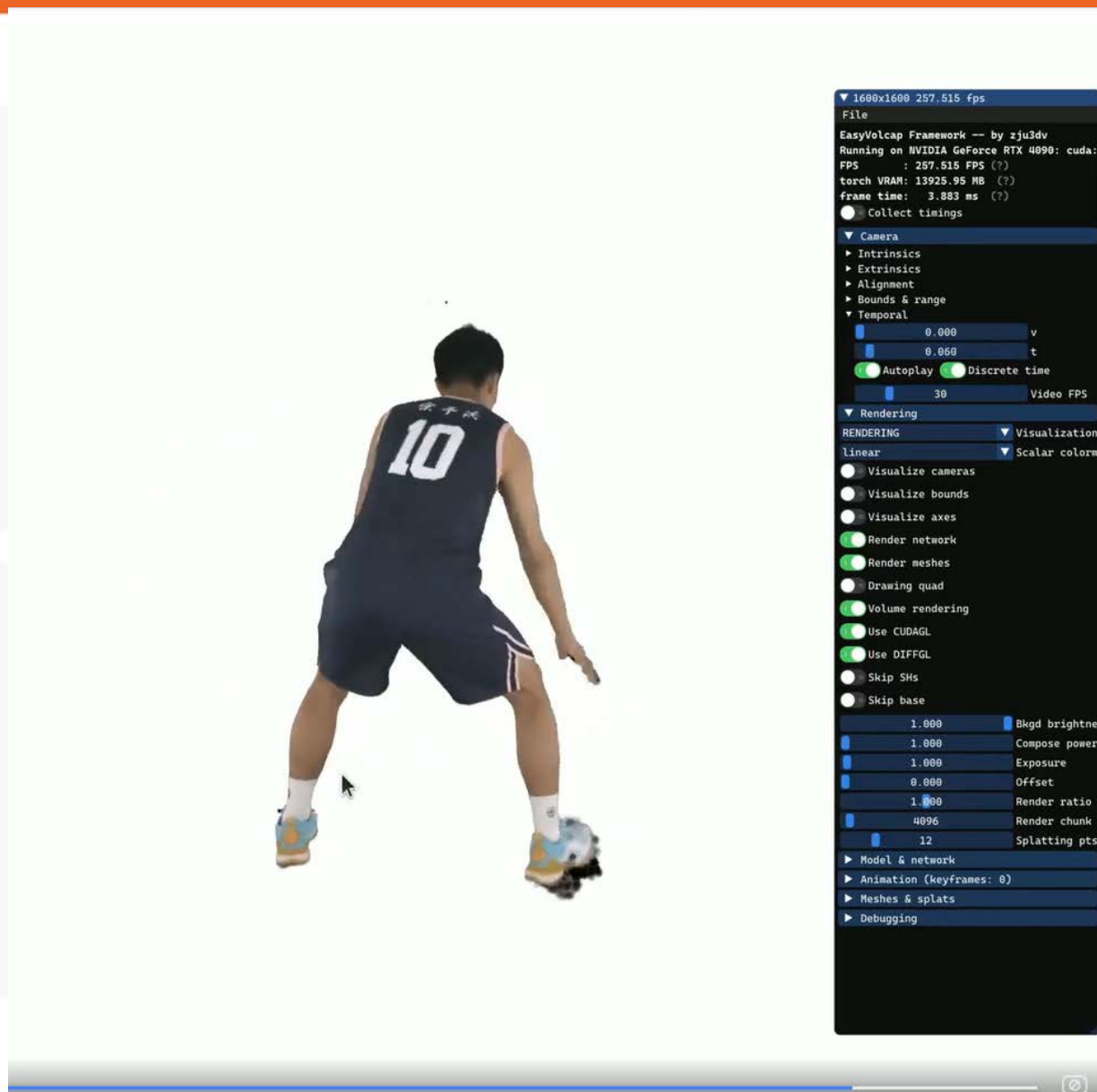














Special effects: shape and motion capture



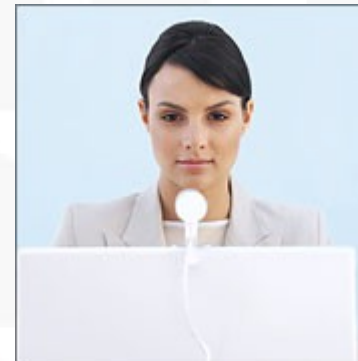


- Many new digital cameras now detect faces
 - Canon, Sony, Fuji, ...

Source: S. Seitz



Fingerprint scanners on many
new laptops,
other devices

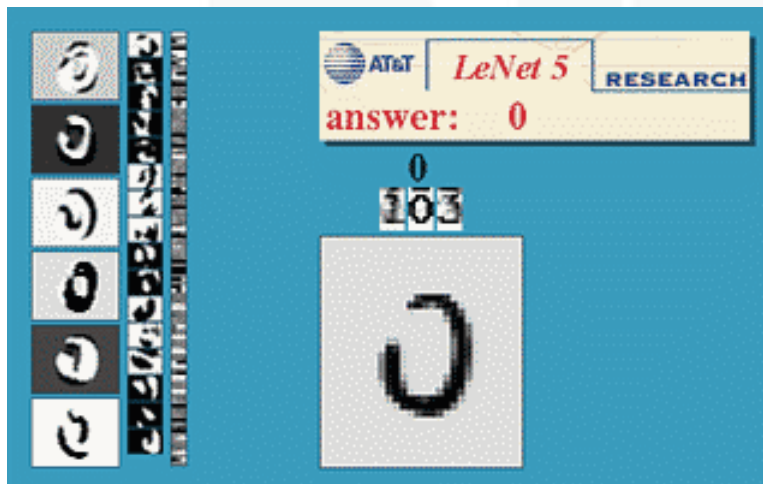


Face recognition systems now beginning to appear
more widely <http://www.sensiblevision.com/>

Source: S. Seitz

Technology to convert scanned docs to text

- If you have a scanner, it probably came with OCR software



Digit recognition, AT&T labs



License plate readers

http://en.wikipedia.org/wiki/Automatic_number_plate_recognition



[NASA'S Mars Exploration Rover Spirit](#) captured this westward view from atop a low plateau where Spirit spent the closing months of 2007.

Vision systems (JPL) used for several tasks

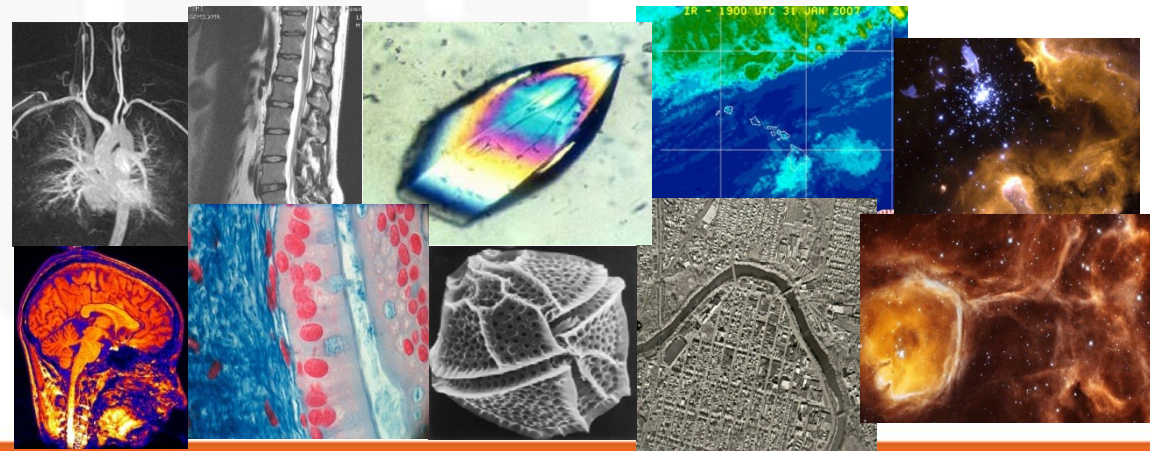
- Panorama stitching
- 3D terrain modeling
- Obstacle detection, position tracking
- For more, read “[Computer Vision on Mars](#)” by Matthies et al.

IT

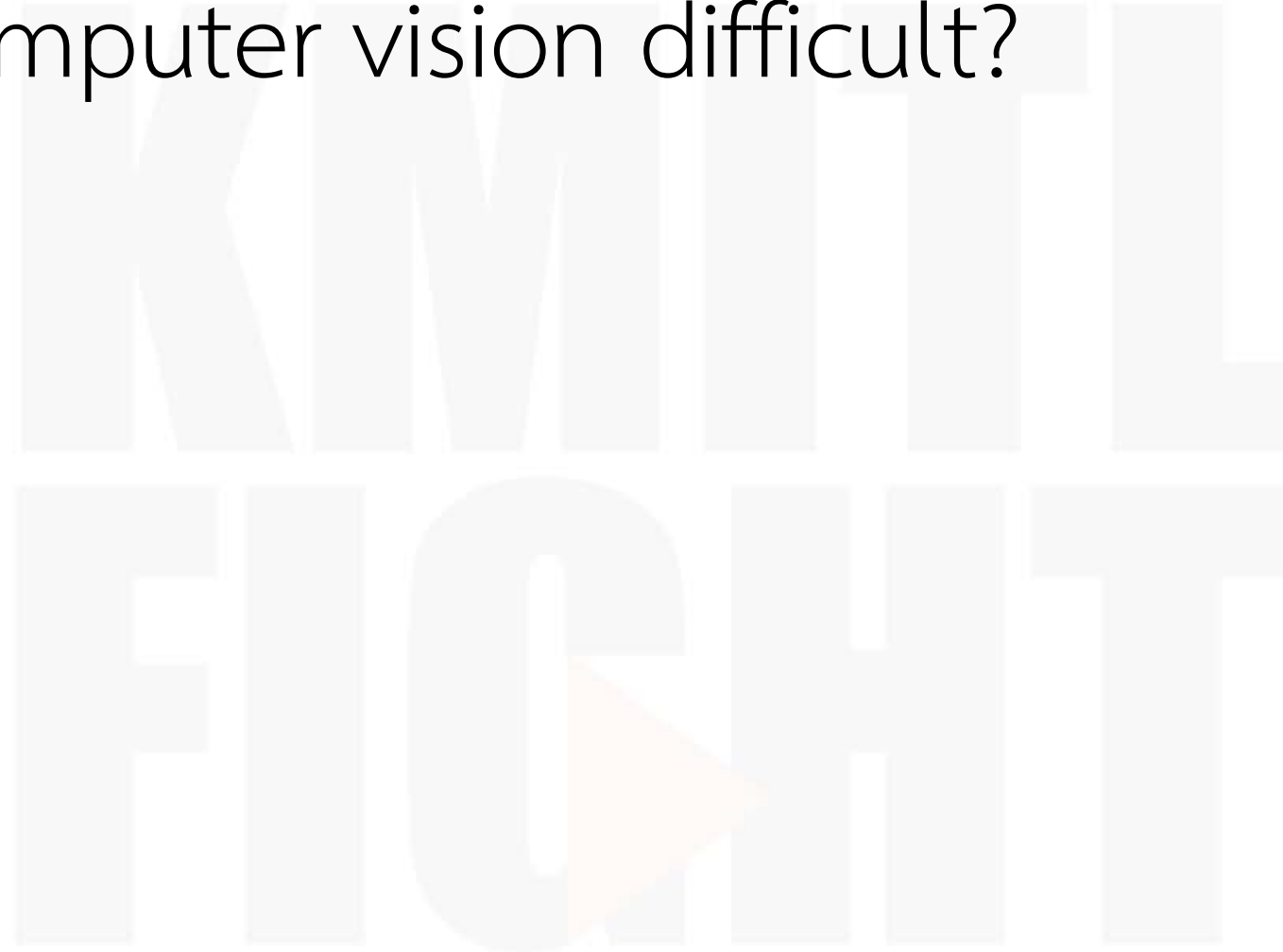
KMITL
สถาบันเทคโนโลยี
พระจอมเกล้า
เจ้าคุณทหารลาดกระบัง

KMITL
FIGHT

-
- A collage of 12 images arranged in a 3x4 grid. The images include: a baby in a white onesie; a couple in wedding attire with a horse; a woman in a blue tank top running; a man in a suit with a nuclear symbol and question mark; a football player in a red jersey; a man in a white shirt holding a large fish; a group of graduates in black caps and gowns; a man in a green shirt and a small alien; a man in a white shirt and sunglasses; a man in a suit with a 'Medical Coverage America' sign; and a baseball player in a white jersey sliding into a base.



Why is computer vision difficult?



Challenges: viewpoint variation



Michelangelo 1475-1564



slide credit: Fei-Fei, Fergus & Torralba

Challenges: illumination

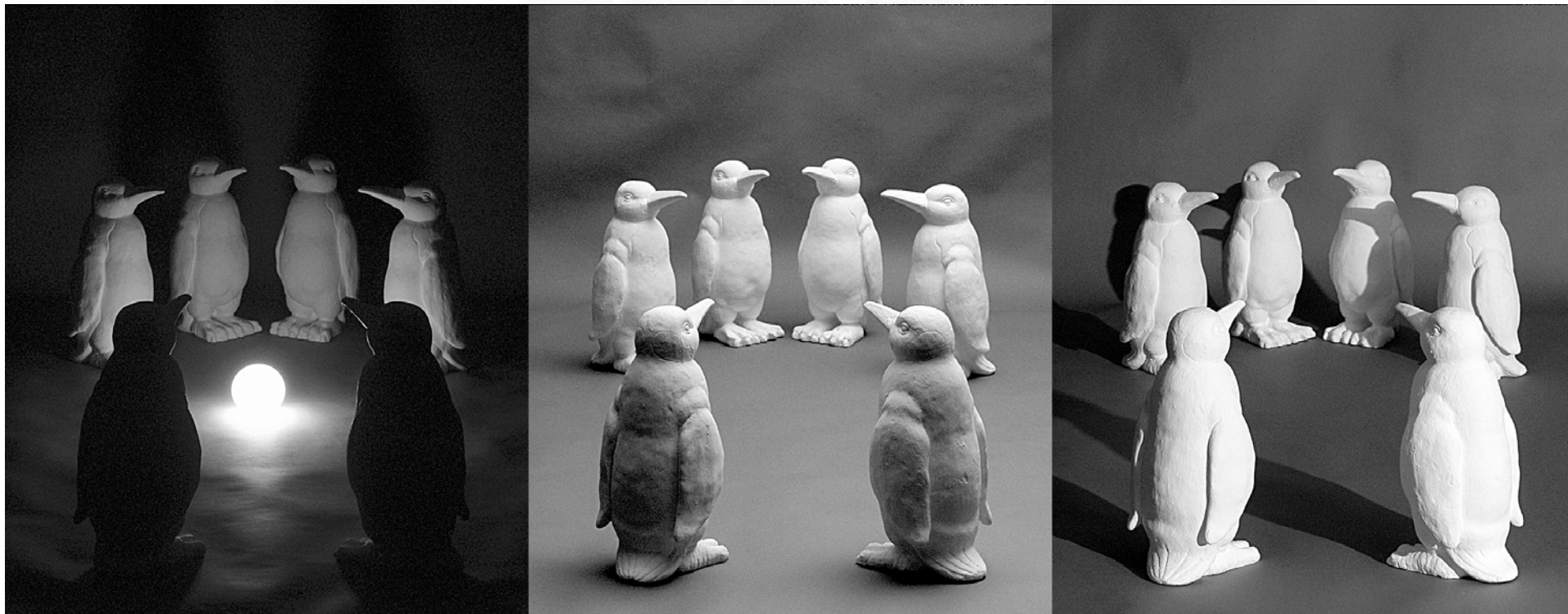


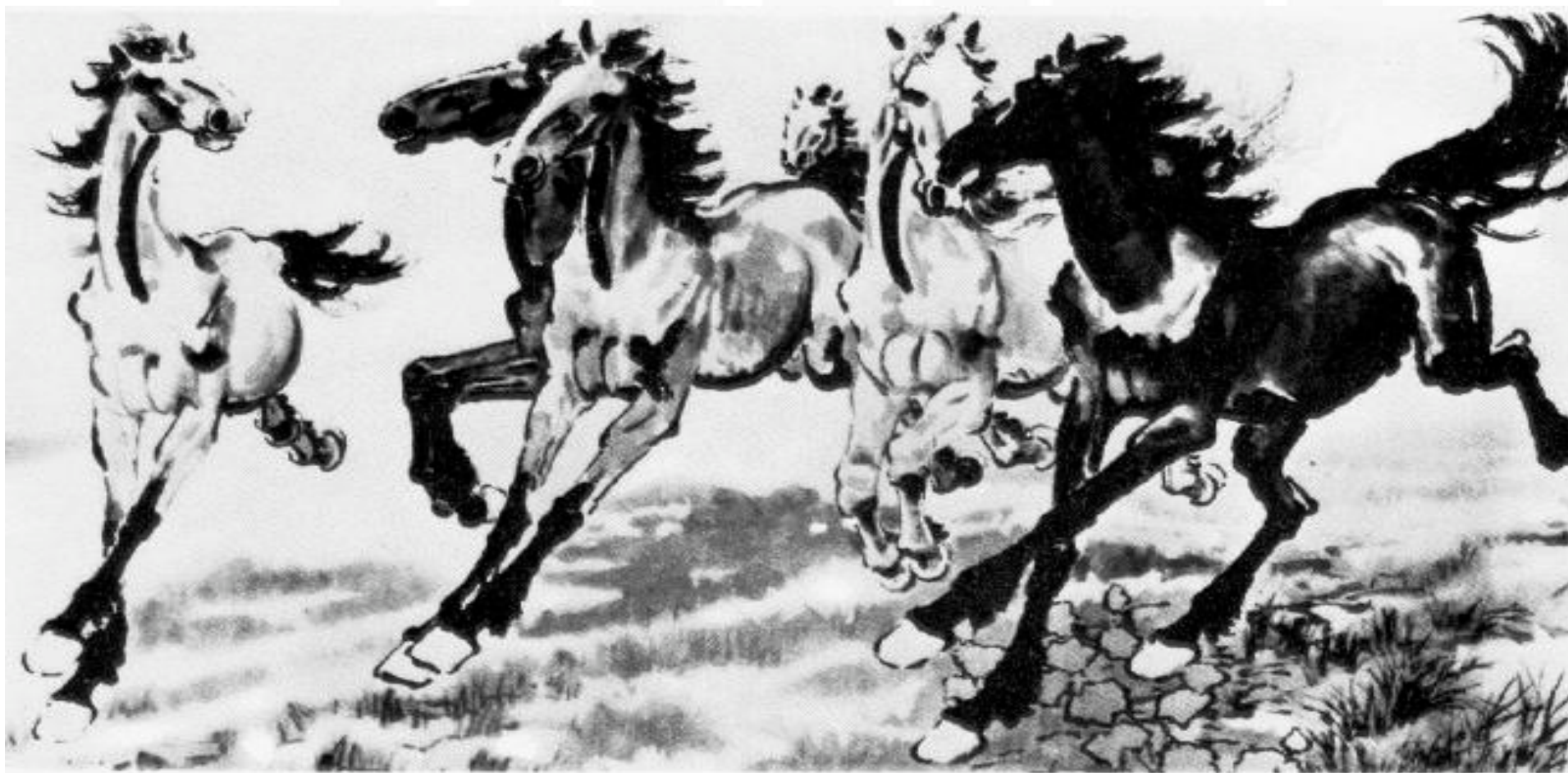
image credit: J. Koenderink

Challenges: scale



slide credit: Fei-Fei, Fergus & Torralba

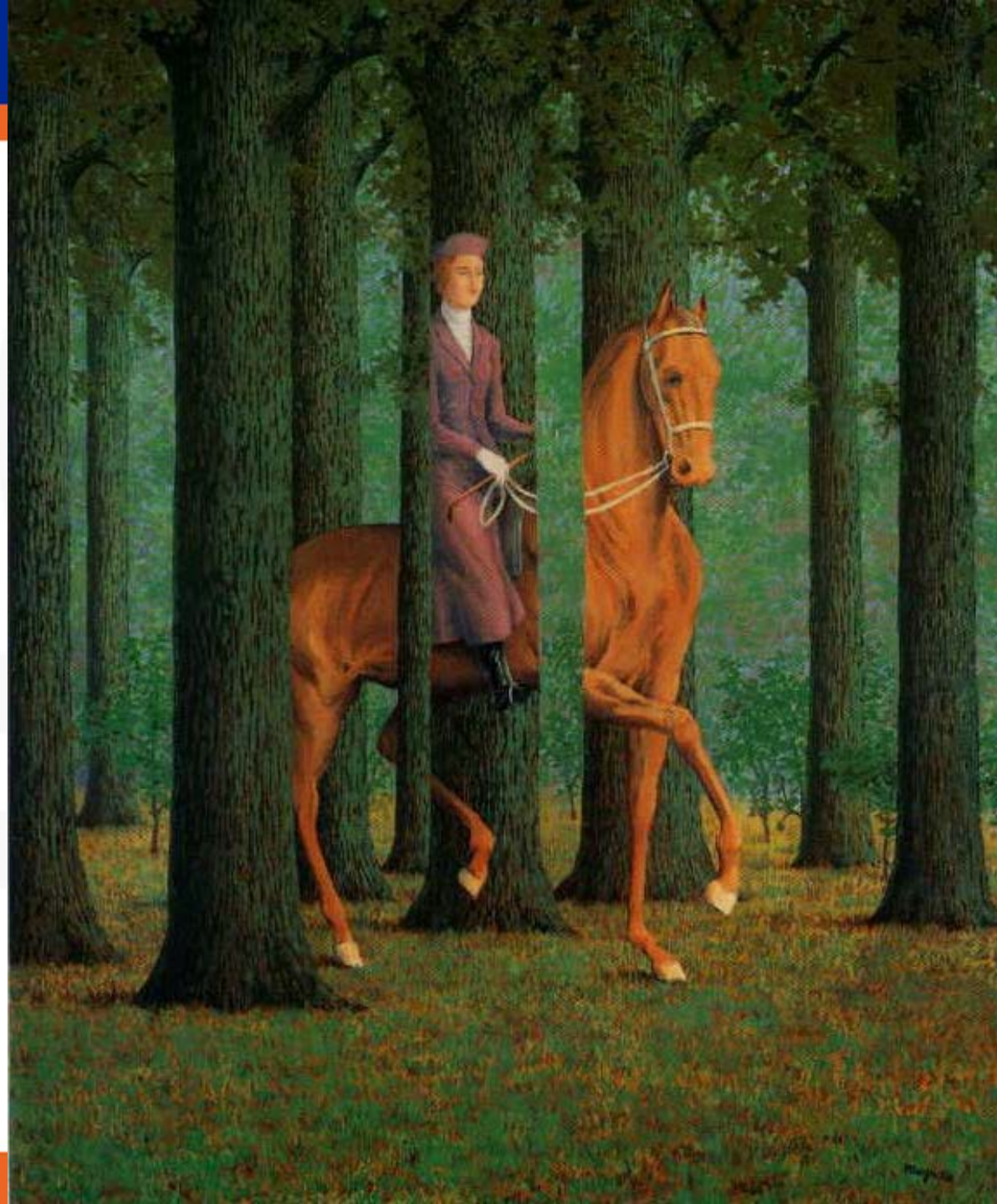
Challenges: deformation



Xu, Beihong 1943

slide credit: Fei-Fei, Fergus & Torralba

Challenges: occlusion



Magritte, 1957

Challenges: background clutter





slide credit: Svetlana Lazebnik

Depth cues: Linear perspective



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- Perception is an inherently ambiguous problem
 - Many different 3D scenes could have given rise to a particular 2D picture



Challenges or opportunities?

- Images are confusing, but they also reveal the structure of the world through numerous cues
- Our job is to interpret the cues!



Image source: J. Koenderink

- Autonomous vehicles & ADAS — รถยนต์ขับเคลื่อนอัตโนมัติและระบบช่วยขับ
- Medical imaging AI — ช่วยแพทย์วิเคราะห์ภาพ X-ray, MRI, CT
- Face ID & payment — ปลดล็อกและชำระเงินด้วยใบหน้า
- Generative AI — สร้างและแก้ไขภาพ (DALL·E, Stable Diffusion)
- Retail automation — ร้านไร้แคชเชียร์ (เช่น Amazon Go)
- AR / VR & spatial computing — ซ้อนโลกเสมือนเข้ากับโลกจริง

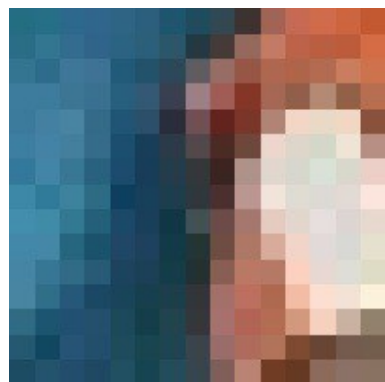
How computer vision do?

Image filtering

Image transformations

Camera matrices

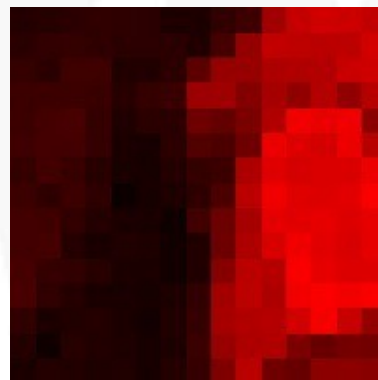
What is an image?



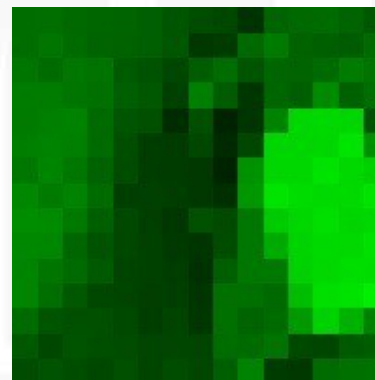
color image patch

How many bits are the
intensity values?

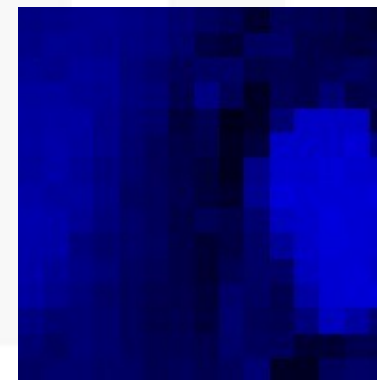
red



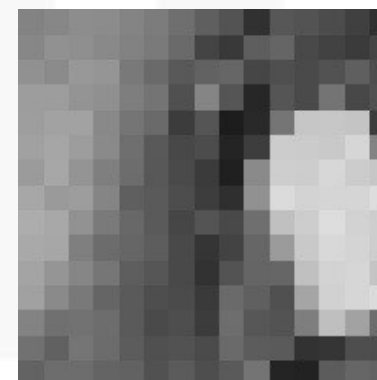
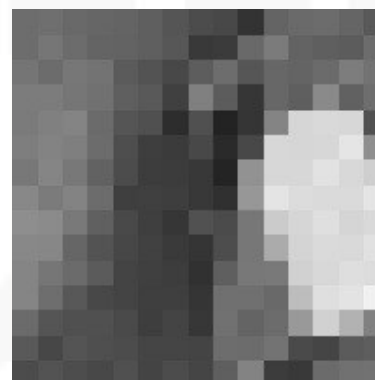
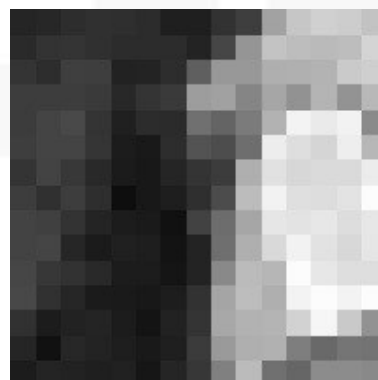
green



blue

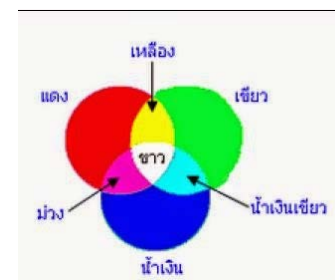


colorized for visualization



actual intensity values per channel

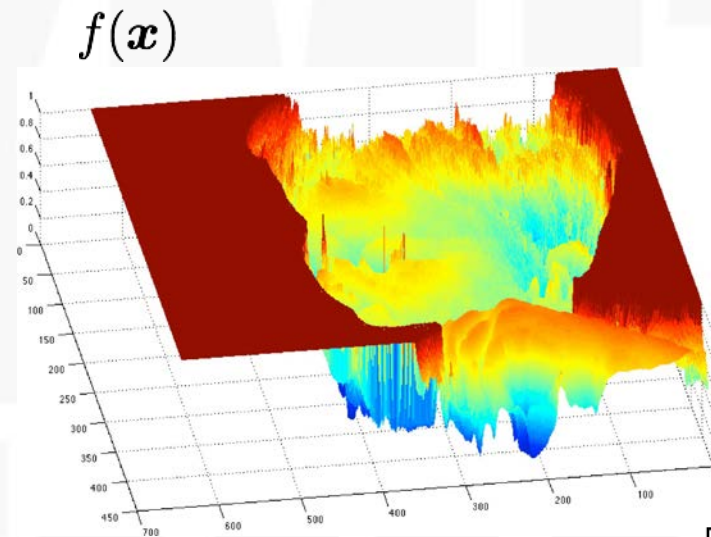
Each channel is a
2D array of
numbers.



What is an image?

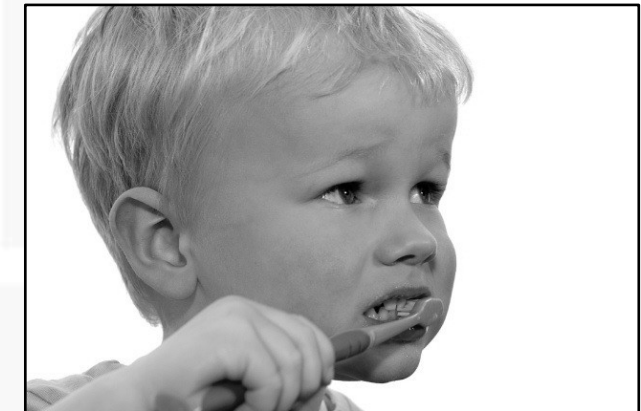


original



domain

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$



grayscale image

What is the range of the image function f ?

Ex.. $\mathbf{x}(R,G,B) = 0.3 \cdot R + 0.59 \cdot G + 0.11 \cdot B$
 $= (\text{Img}[i,j,0] \cdot 0.3 + \text{Img}[i,j,1] \cdot 0.59 + \text{Img}[i,j,2] \cdot 0.11)$

11	255	5	55	101	25	→	17	111
1	219	15	211	12	19		??	??

$$\mathbf{x}(R,G,B) = \text{Floor}[0.3 \cdot 11 + 0.59 \cdot 5 + 0.11 \cdot 101] = 17$$

$$\mathbf{x}(R,G,B) = \text{Floor}[0.3 \cdot 255 + 0.59 \cdot 55 + 0.11 \cdot 25] = 111$$

What is an image?



grayscale image

$$\text{Grayscale} = \begin{cases} 255 & ; \text{if grayscale} > 127 \\ 0 & ; \text{else} \end{cases}$$



Black and white image

Image filtering



changes pixel values

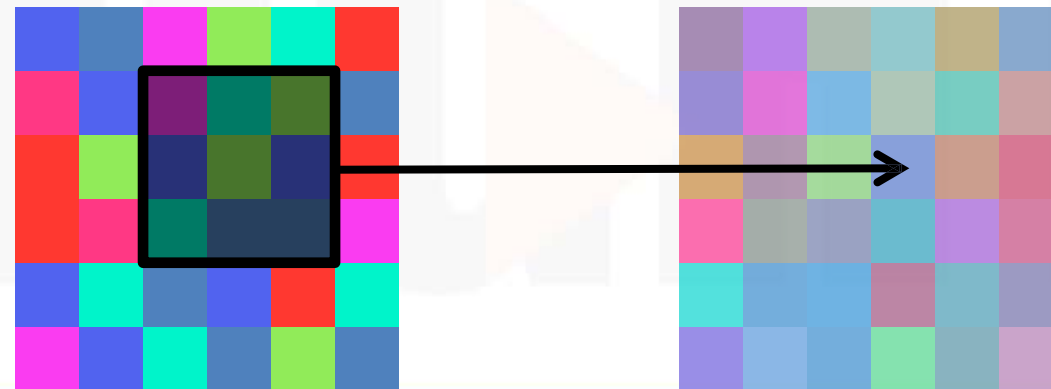
What types of image filtering can we do?

Point Operation



point processing

Neighborhood Operation



“filtering”

Examples of point processing

How would you implement these?

original



$$x$$

darken



0 51 102 153 204 255

lower contrast



$$\frac{x}{2}$$

non-linear lower contrast



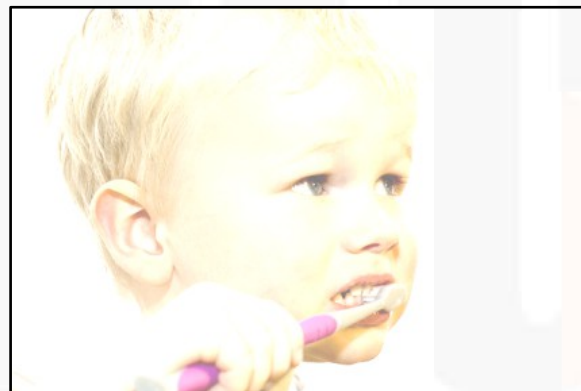
$$\left(\frac{x}{255}\right)^{1/3} \times 255$$

invert



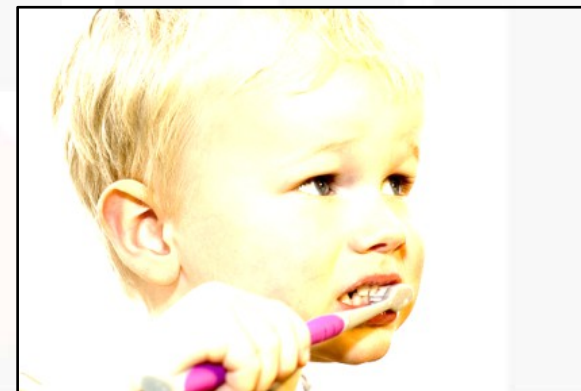
$$255 - x$$

lighten



$$x + 128$$

raise contrast



$$x \times 2$$

non-linear raise contrast



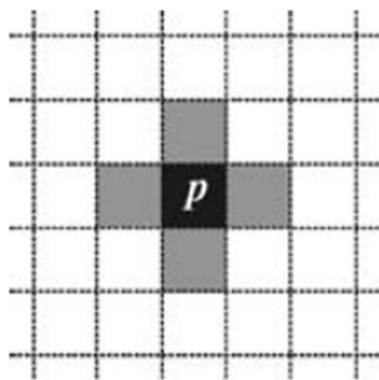
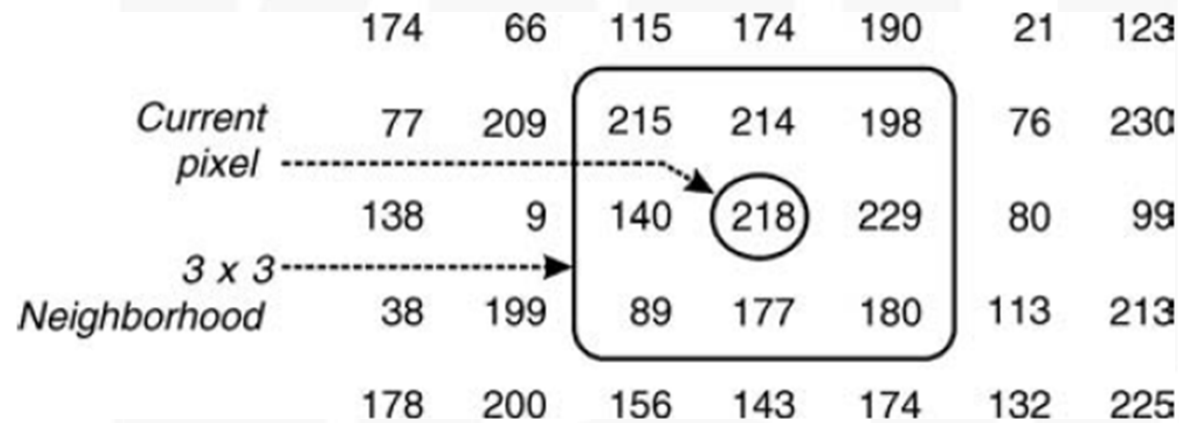
$$\left(\frac{x}{255}\right)^2 \times 255$$

Neighborhood processing

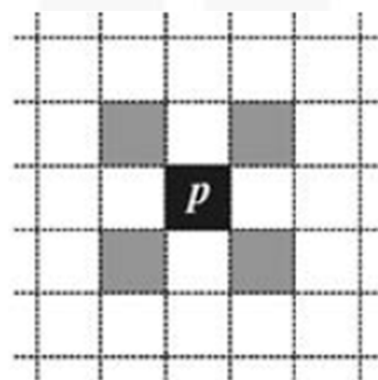
The process of moving the center point creates new neighborhoods, one for each pixel in the input image.

- The two principal terms used to identify this operation are *neighborhood processing* and *spatial filtering*.
- If the computation performed on the pixels of the neighborhoods are linear,
 - the operation is called *linear spatial filtering* (the term *spatial convolution* also used);
 - otherwise, it is called *nonlinear spatial filtering*.

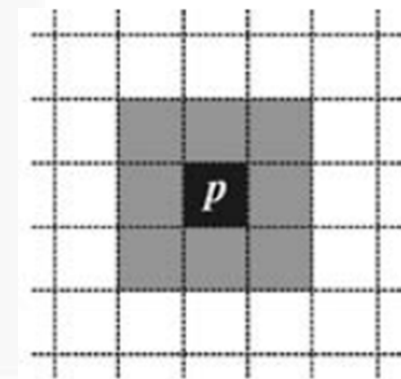
BASIC TERMINOLOGY



(a) 4-neighborhood;



(b) diagonal neighborhood;

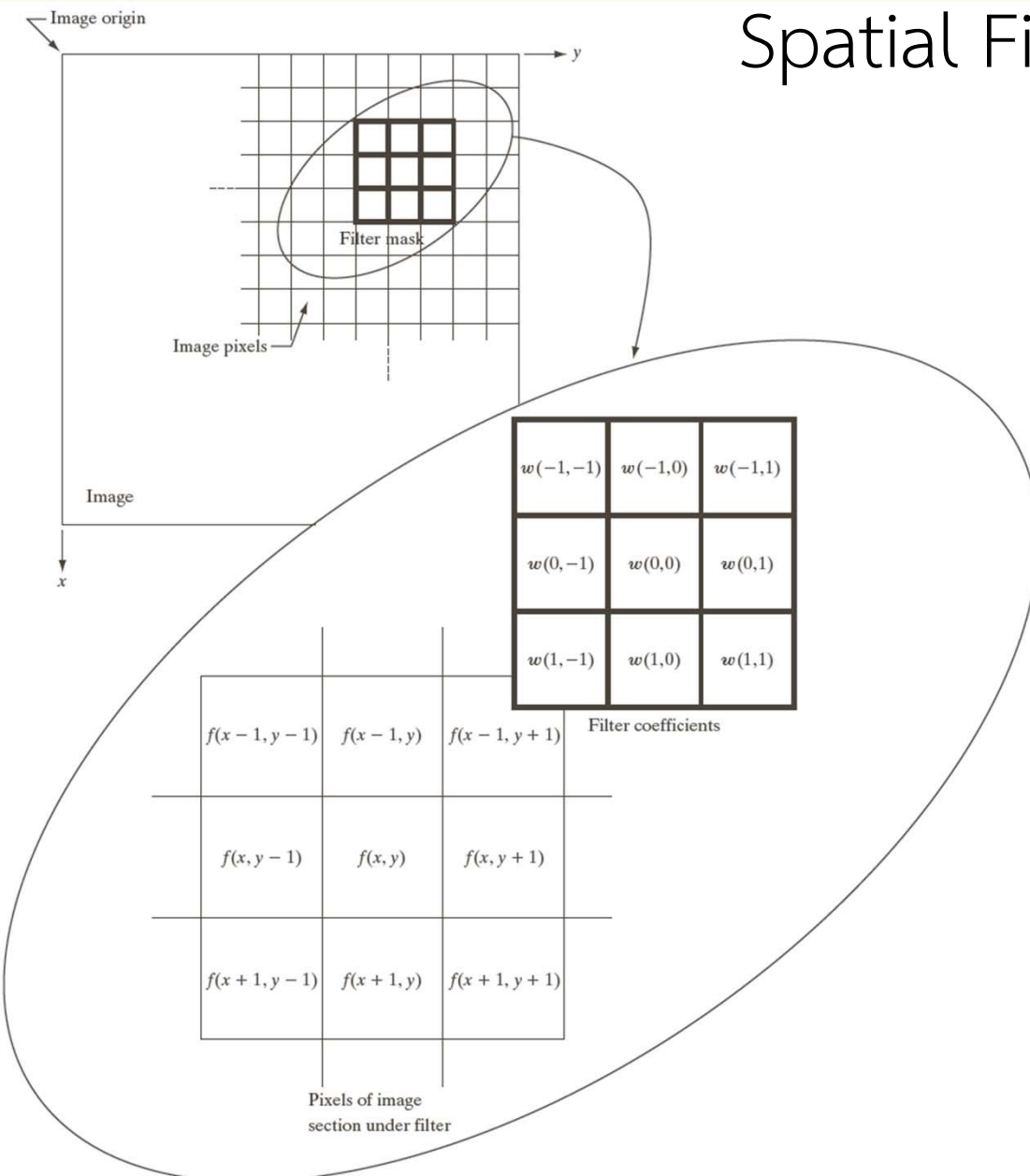


(c) 8-neighborhood.

Spatial Filtering

Spatial filtering is a neighborhood processing consisting of:

1. defining a **center point**, (x, y) ;
2. performing an operation that involves only the pixels in a **predefined neighborhood** of **the center point**;
3. letting the result of that operation be the **"response"** of the process at that point;
4. repeating the process for every point in the image.



CONVOLUTION

- In one-dimensional

$$A * B = \sum_{j=-\infty}^{\infty} A(j) \cdot B(x - j)$$

Ex: Let $A = \{0, 1, 2, 3, 2, 1, 0\}$, and $B = \{1, 3, -1\}$.

1. $A(x=1)$

A		0	1	2	3	2	1	0
B	-1	3	1					
$A * B$		1						



$$(0 \times (-1)) + (0 \times 3) + (1 \times 1) = 1$$

CONVOLUTION (cont.)

2. $A(x=2)$

A	0	1	2	3	2	1	0
B	-1	3	1				
$A * B$	1	5					



$$(0 \times (-1)) + (1 \times 3) + (2 \times 1) = 5$$

⋮

7. $A(x=7)$

A	0	1	2	3	2	1	0	
B						-1	3	1
$A * B$	1	5	8	8	4	1	-1	



$$(1 \times (-1)) + (0 \times 3) + (0 \times 1) = -1$$

2D Convolution

$$g(x, y) = \sum_{k = -m/2}^{m/2} \sum_{l = -n/2}^{n/2} w(k, l) f(x - k, y - l)$$

EXAMPLE

120	201	100
50	25	45
10	20	56

Image

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

Filter

new value at position (2,2):

$$\text{new value} = ((120+201+100+50+25+45+10+20+56)/9) = 69.6$$

updated image matrix given by :

120	201	100
50	69.6	45
10	20	56

Image filtering

Filtering operations use masks as the *spatial filters*, which are named based on their behavior in the *spatial frequency*:

- *Low-pass filters* (LPFs) preserve low-frequency components (i.e., coarser details and homogeneous areas in the image).
- *High-pass filters* (HPFs) enhance high-frequency components (i.e., fine details in the image)

Image filtering



Original image



LPF image



HPF image

Spatial Averaging and Spatial Low-Pass Filtering

$$g(x, y) = \sum_{k=1}^n \sum_{l=1}^n w(k, l) f(x - k, y - l)$$

where g and f are the output and input images.

w is the weight matrix of the size $n \times n$

For average filter, all elements in w are 1, so that

$$g(x, y) = \frac{1}{N_w} \sum_{k=1}^n \sum_{l=1}^n f(x - k, y - l)$$

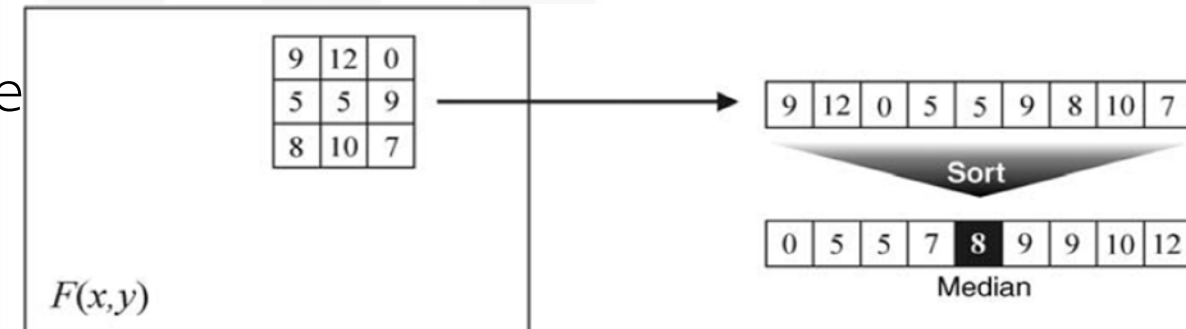
where $w = 1/N_w$ and N_w is the number of elements in w

Median filter

The median filter is a **nonlinear filter** used in image processing. It works by:

- **sorting the pixel values** within a neighborhood,
- **finding the median value**, and
- replacing the original pixel value with the median.

Median filtering process



$$g(x, y) = \text{median}\{f(x - k, y - l), (k, l) \in W\}$$

Original



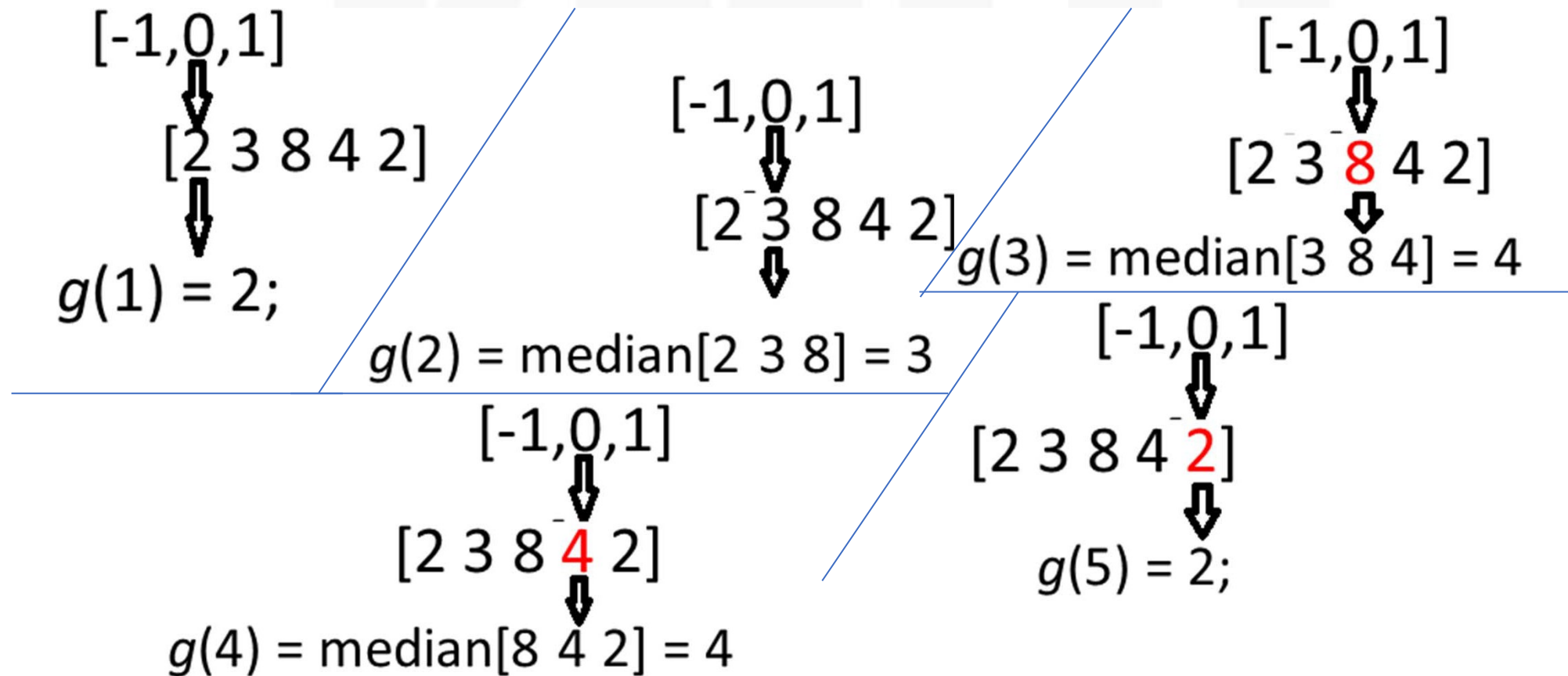
Median,3x3



Median,5x5



Ex. Let $f(x) = [2 \ 3 \ 8 \ 4 \ 2]$, $W = [-1, 0, 1]$, the filtering operation can be performed by:



The output of median filter is $g(x) = [2 \ 3 \ 4 \ 4 \ 2]$.

Applications

Neighborhood operations is core concept of commonly used pre-processing techniques from blurring to edge detection, some applications are listed below:

Mean Filtering

- Application: Image Smoothing
- Example: Removing noise from a photograph to improve visual aesthetics.

Median Filtering

- Application: Salt-and-pepper Noise Removal
- Example: Cleaning medical images where outliers may represent errors.

Gaussian Filtering

- Application: Blurring for Noise Reduction
- Example: Enhancing fingerprint images by reducing high-frequency noise.

Laplacian Filter

- Application: Edge Detection
- Example: Identifying boundaries between objects in medical imaging.

Sobel and Prewitt Operators

- Application: Gradient-Based Edge Detection
- Example: Highlighting edges in satellite imagery for feature extraction.

How computer vision do?

Image filtering

Image transformations

Camera matrices

What types of image transformations

can we do?

F



Filtering

$$\downarrow G(\mathbf{x}) = h\{F(\mathbf{x})\}$$

G



changes *range* of image function

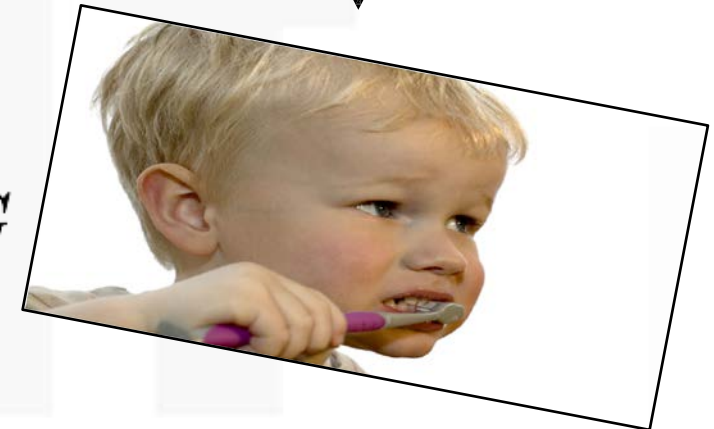
F



Warping

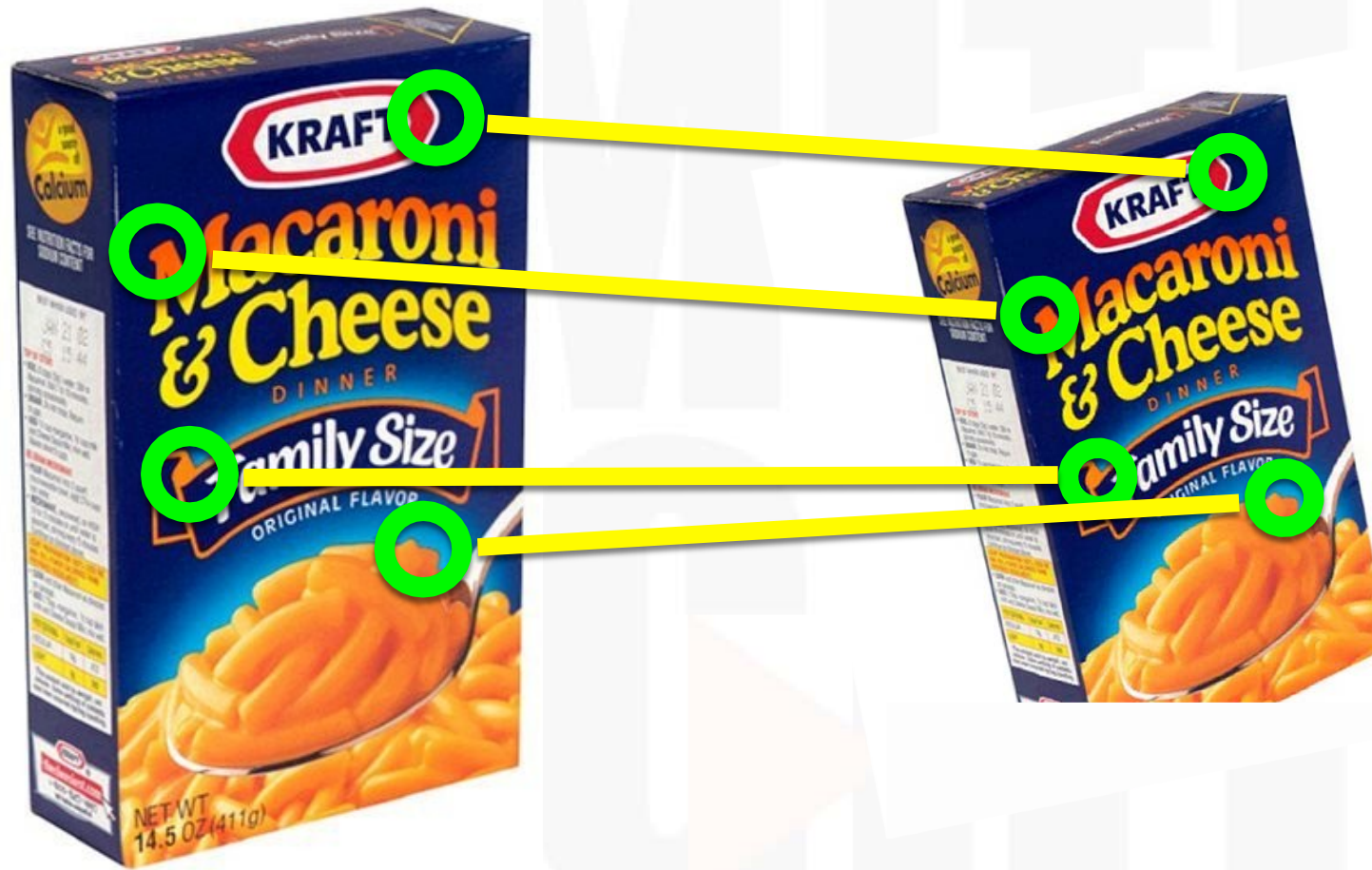
$$\downarrow G(\mathbf{x}) = F(h\{\mathbf{x}\})$$

G



changes *domain* of image function

Warping example: feature matching



- object recognition
- 3D reconstruction
- augmented reality
- image stitching

How do you compute the transformation?

Warping example: feature matching

Given a set of matched feature points:

$$\{x_i, x'_i\}$$

point in one
image

point in the
other image

and a transformation:

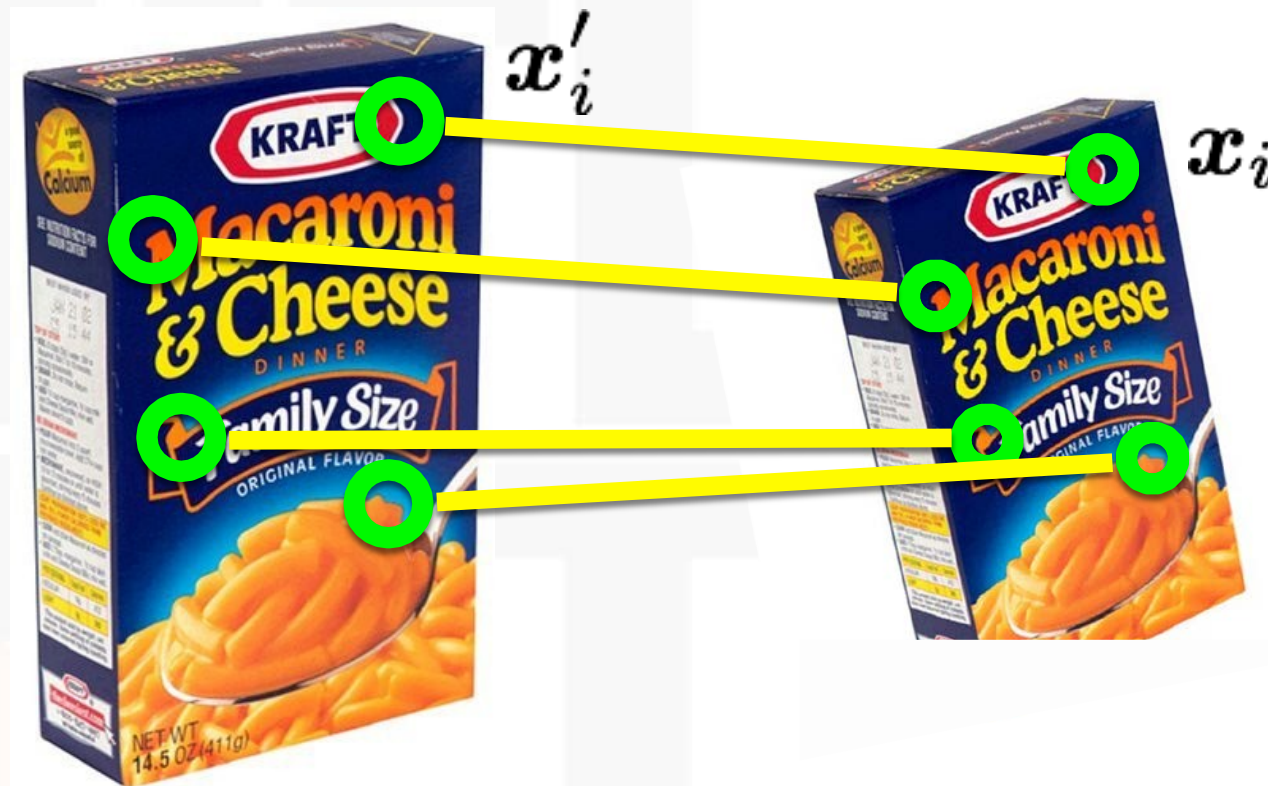
$$x' = f(x; p)$$

transformation
function

parameters

find the best estimate of the parameters

p



What kind of transformation functions

f are there?

2D transformations



translation



rotation



aspect



affine



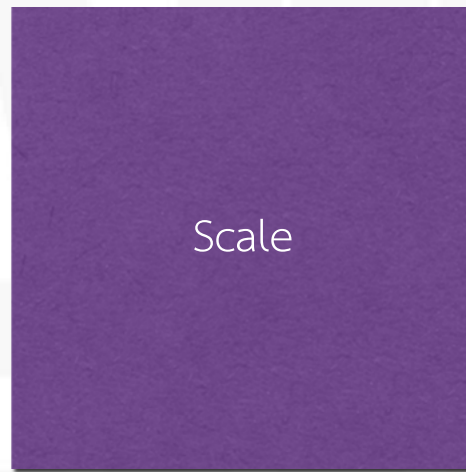
perspective



cylindrical

2D planar transformations

y



Scale

$$x' = ax$$

$$y' = by$$

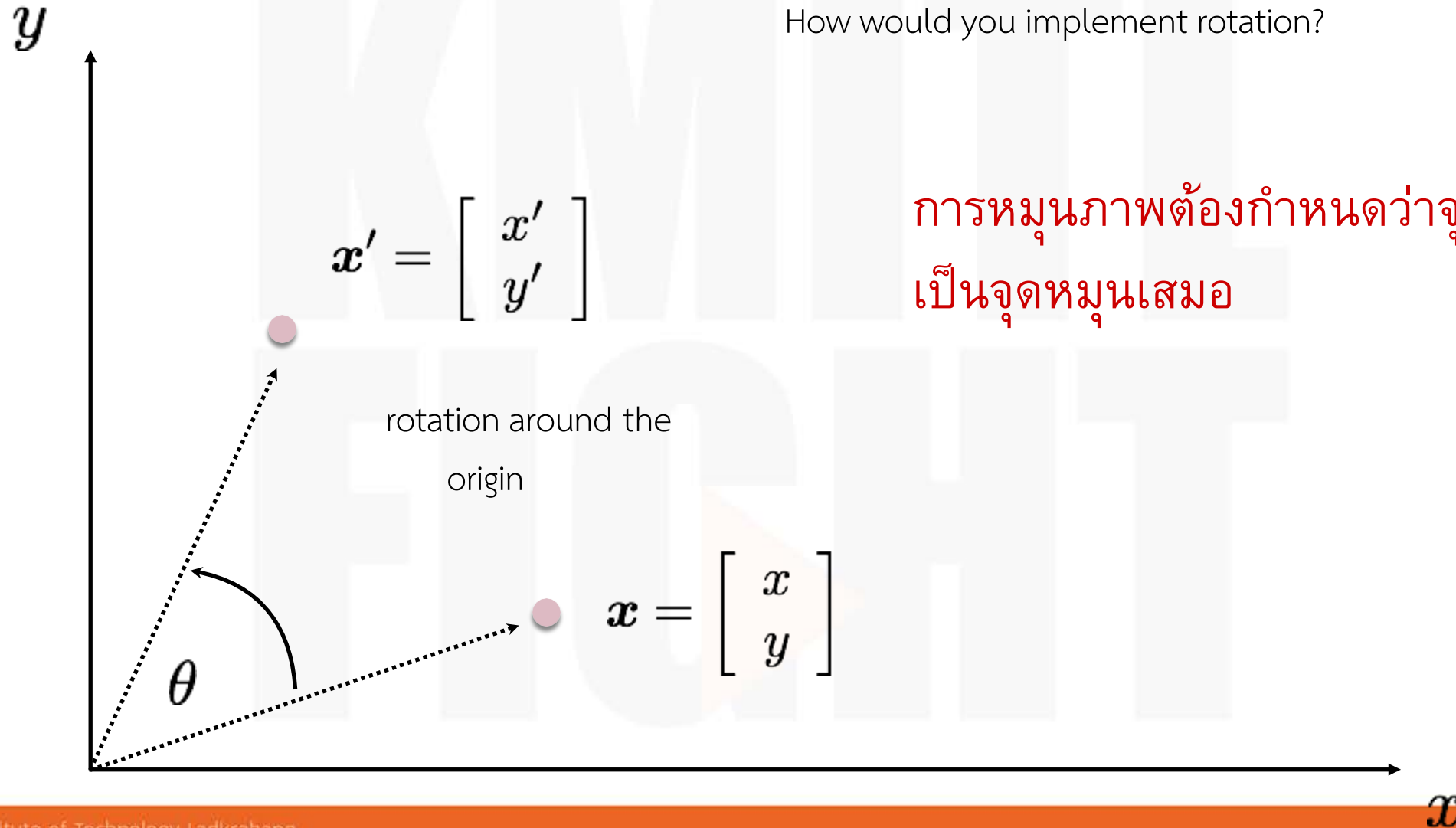
matrix representation of scaling:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \underbrace{\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}}_{\text{scaling matrix } S} \begin{bmatrix} x \\ y \end{bmatrix}$$

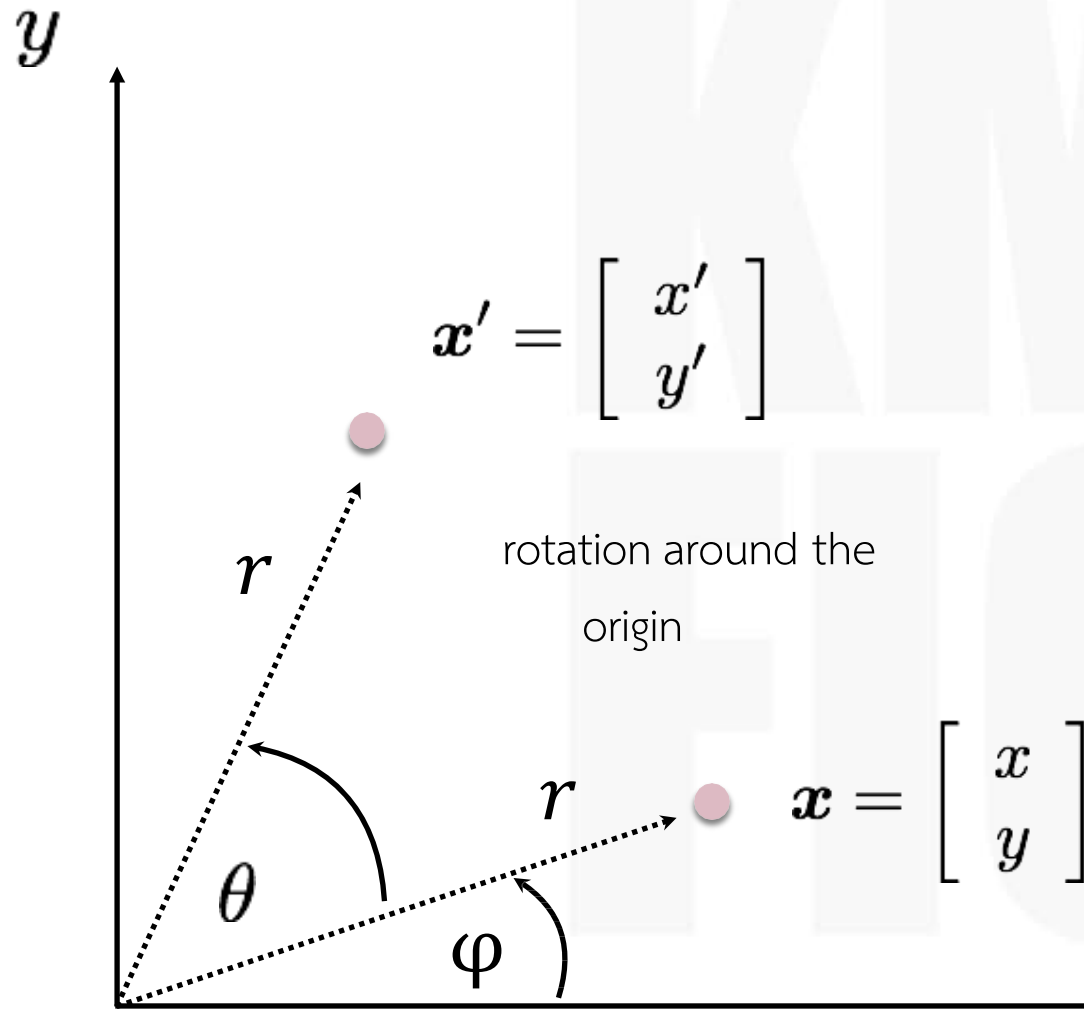
- Each component multiplied by a scalar
- Uniform scaling - same scalar for each component

x

2D planar transformations



2D planar transformations



Polar coordinates

$$x = r \cos(\phi)$$

$$y = r \sin(\phi)$$

$$x' = r \cos(\phi + \theta)$$

$$y' = r \sin(\phi + \theta)$$

Trigonometric Identity...

$$x' = r \cos(\phi) \cos(\theta) - r \sin(\phi) \sin(\theta)$$

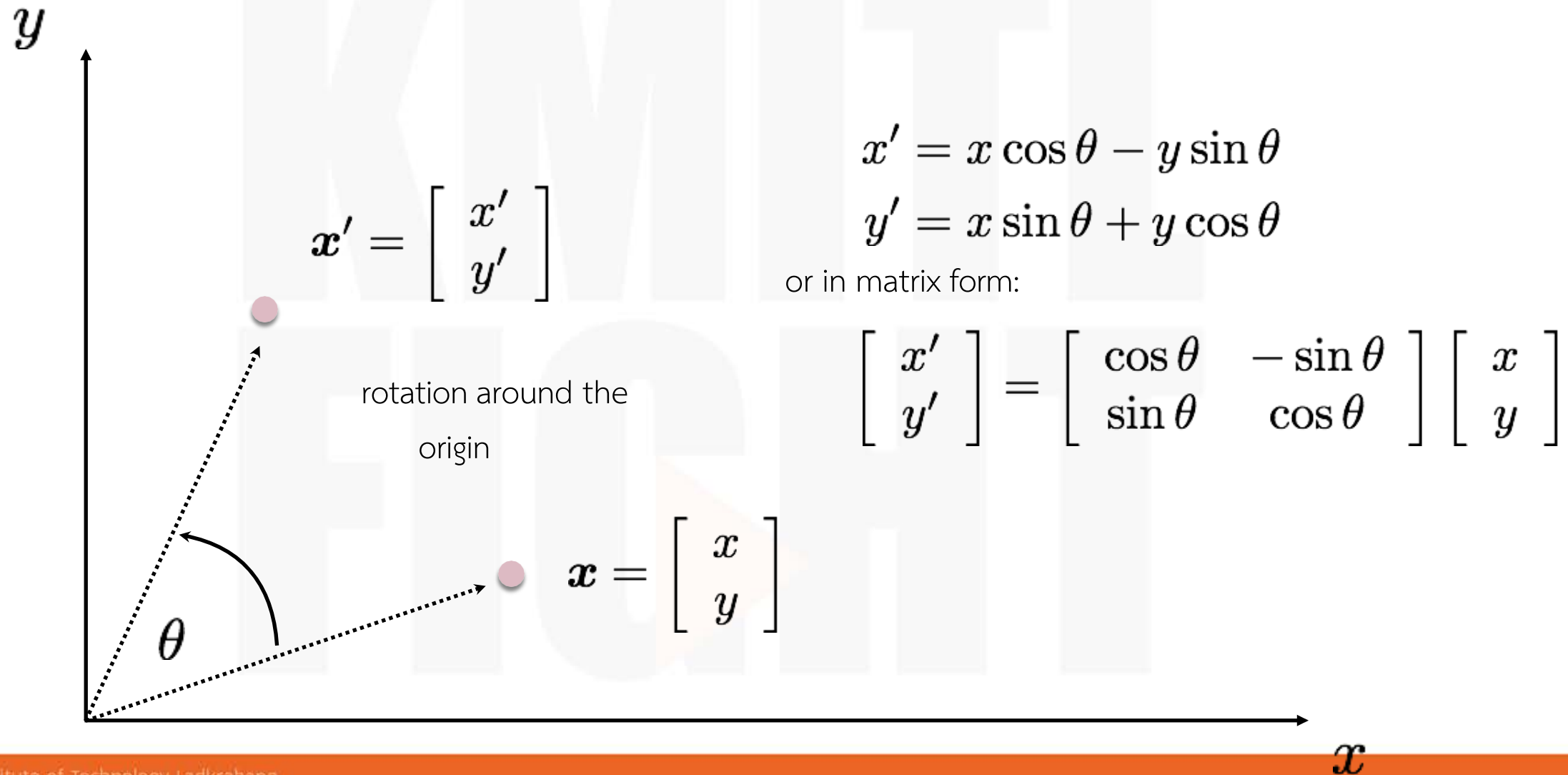
$$y' = r \sin(\phi) \sin(\theta) + r \cos(\phi) \cos(\theta)$$

Substitute...

$$x' = x \cos(\theta) - y \sin(\theta)$$

$$y' = y \sin(\theta) + x \cos(\theta)$$

2D planar transformations



2D planar transformations



Scale

$$\mathbf{M} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

Flip across y

$$\mathbf{M} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Rotate

$$\mathbf{M} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Flip across origin

$$\mathbf{M} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

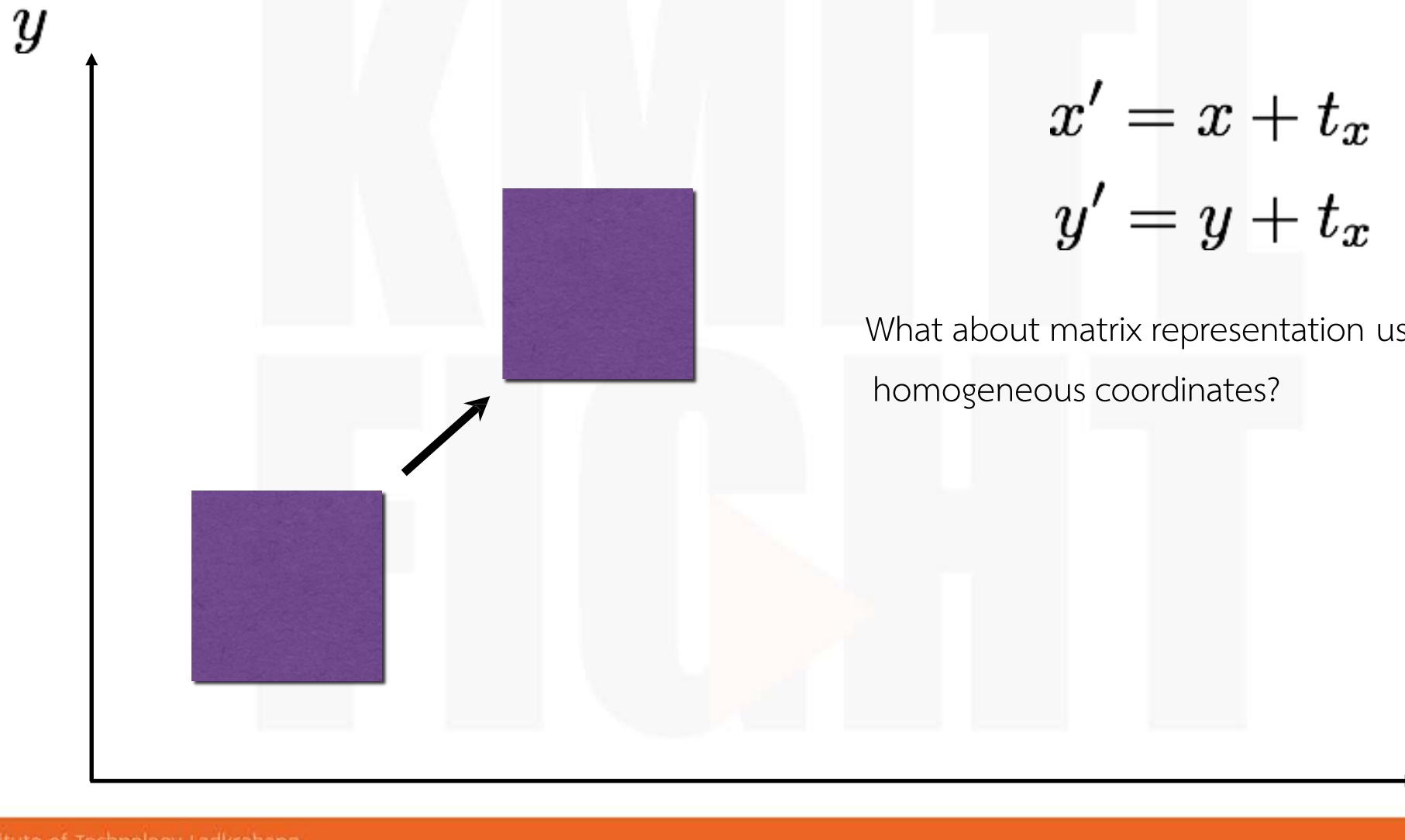
Shear

$$\mathbf{M} = \begin{bmatrix} 1 & s_x \\ s_y & 1 \end{bmatrix}$$

Identity

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

2D translation



Homogeneous coordinates

heterogeneous
coordinates

homogeneous
coordinates

$$\begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

add a 1 here

- Represent 2D point with a 3D vector

Re-write these transformations as 3x3 matrices:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

translation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

scaling

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \Theta & -\sin \Theta & 0 \\ \sin \Theta & \cos \Theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & \beta_x & 0 \\ \beta_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

shearing

Matrix composition

Transformations can be combined by matrix multiplication:

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \left(\begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \Theta & -\sin \Theta & 0 \\ \sin \Theta & \cos \Theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} sx & 0 & 0 \\ 0 & sy & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

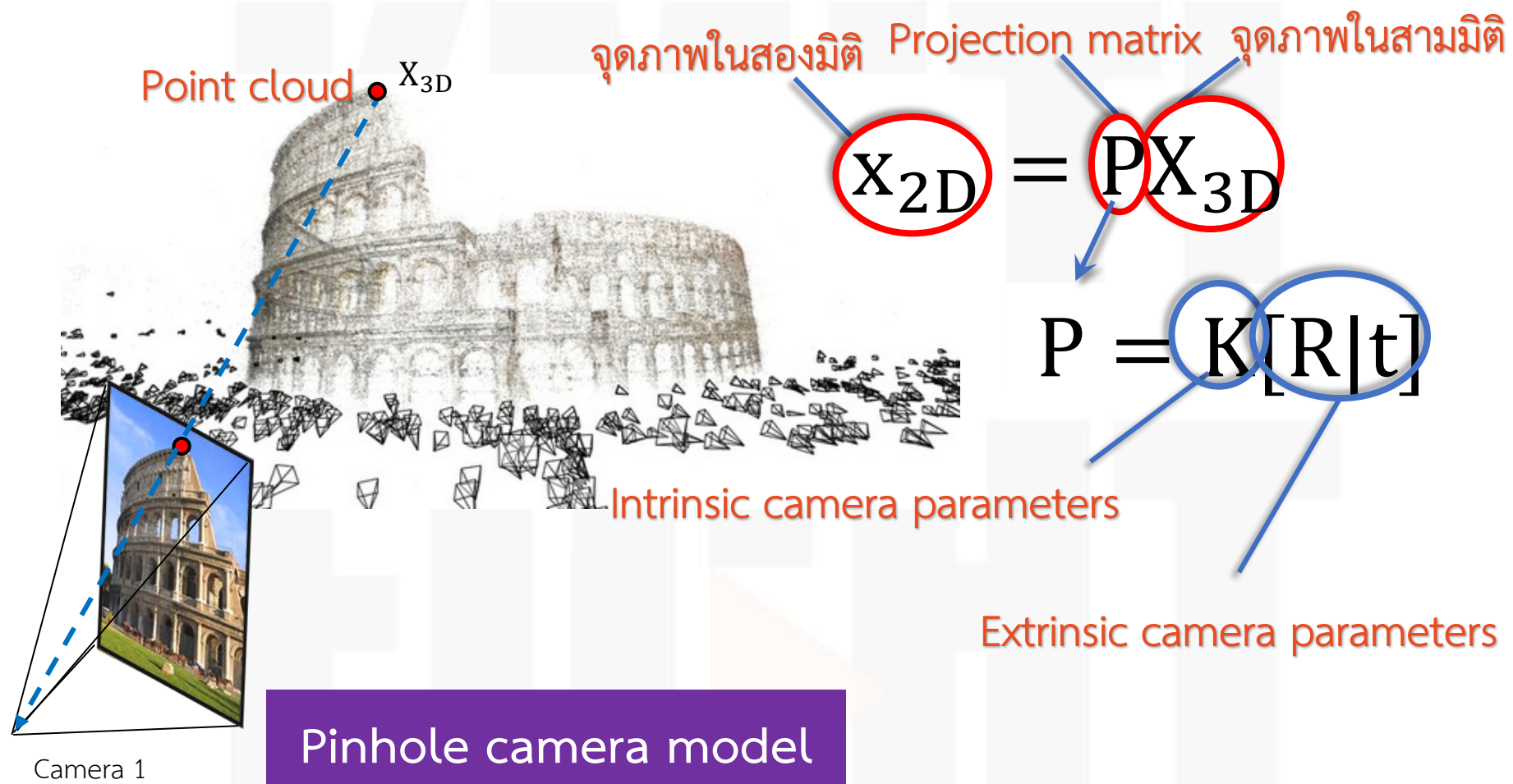
p' = translation(t_x, t_y) rotation(Θ) scale(s, s) p

How computer vision do?

Image filtering

Image transformations

Camera matrices



Forward Projection onto image plane. 3D (X,Y,Z) projected to 2D (x,y)

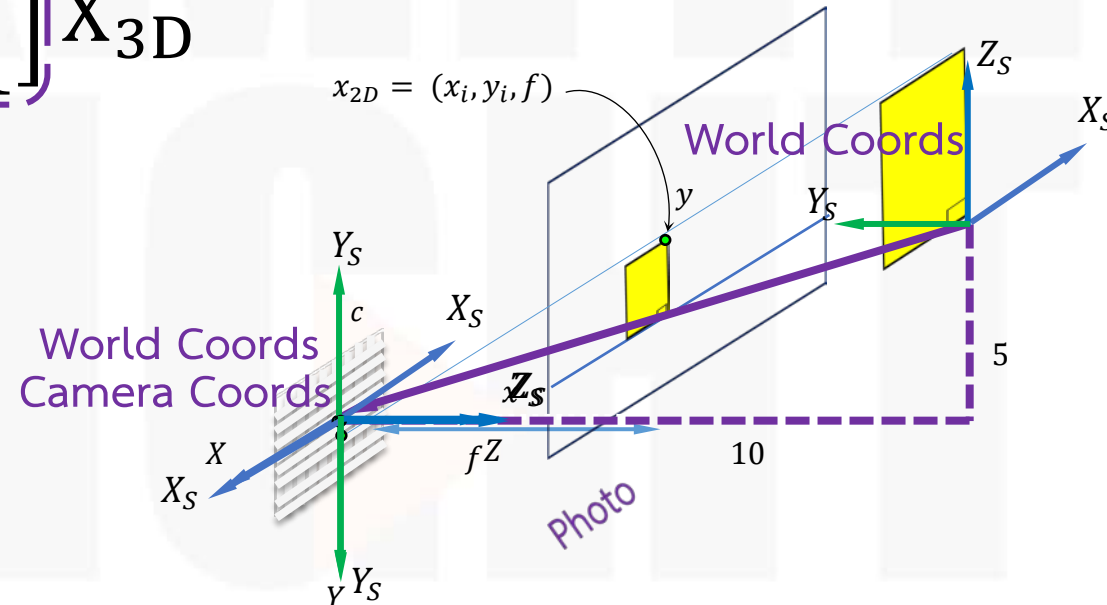
Pinhole camera model

$$x_{2D} = K[R|t]X_{3D}$$

$$x_{2D} = K \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} X_{3D}$$

Extrinsic camera parameters

Camera rotation and translation



Pinhole camera model

Forward Projection onto image plane.

3D (X,Y,Z) projected to 2D (x,y)

$$X_{2D} = K[R|t]X_{3D}$$

Intrinsic camera parameters

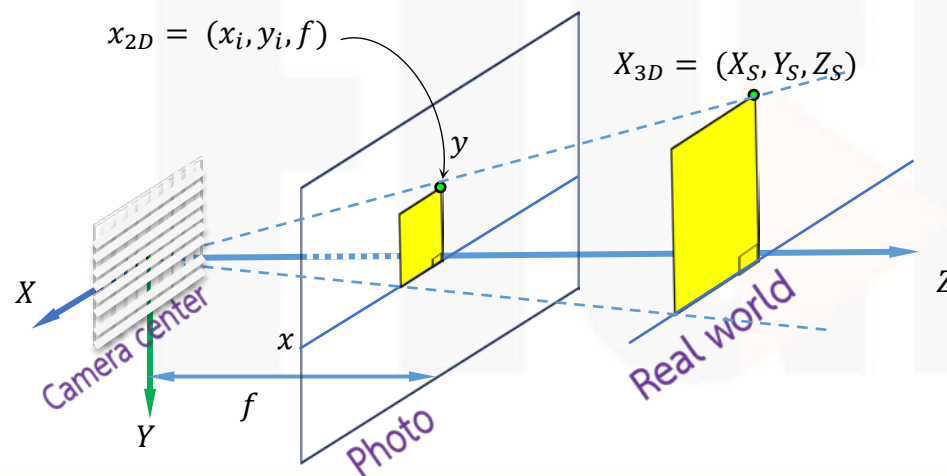
$$K = \begin{bmatrix} fm_x & s & p_x \\ 0 & fm_y & p_y \\ 0 & 0 & 1 \end{bmatrix}$$

f = focal length

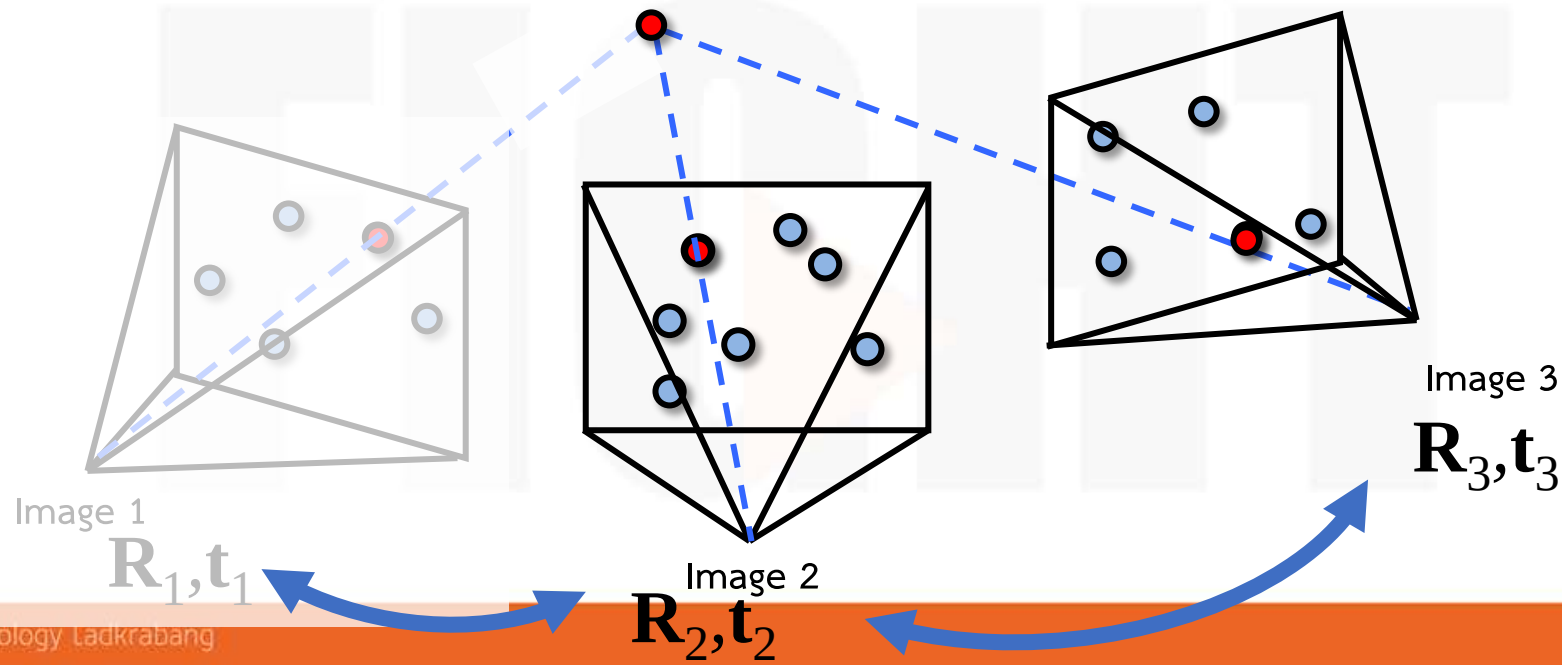
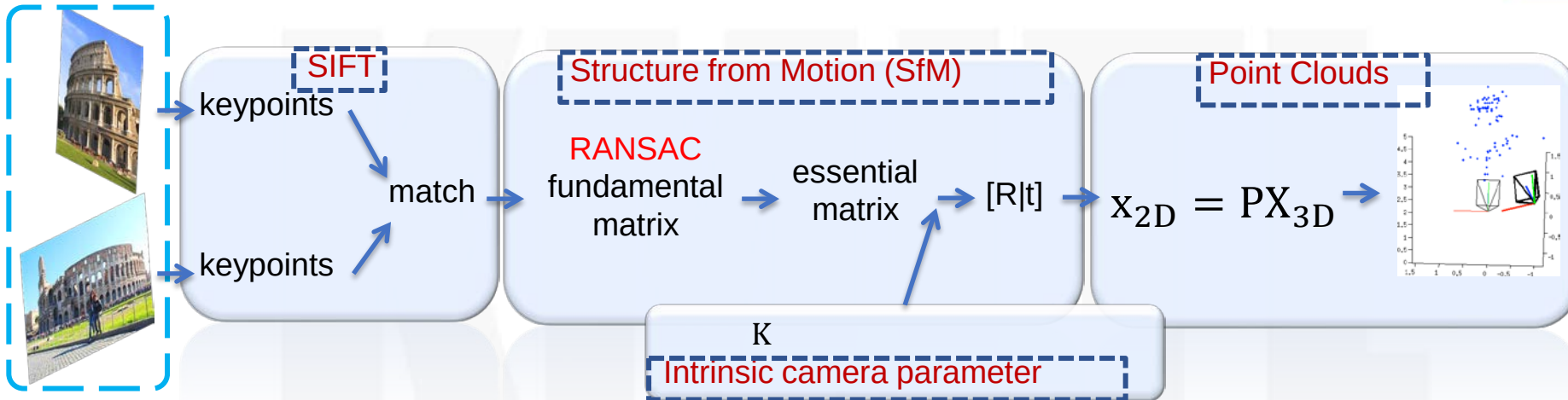
$m_{x,y}$ = camera pixel size

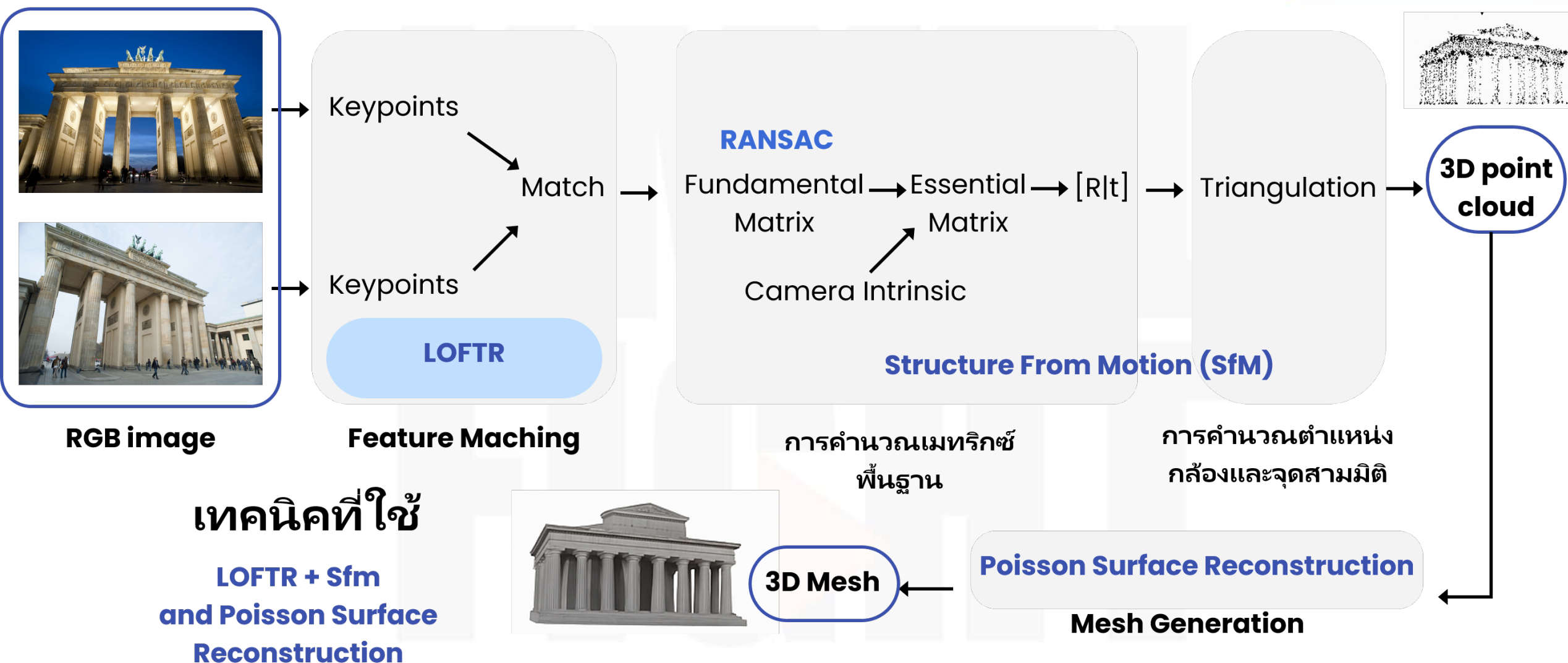
$p_{x,y}$ = principal point coordinates

s = skew parameter



Reconstruct 2D (images) to 3D(point clouds)





Lab Week 2

Computer Vision Preprocessing Toolkit

จำลอง dataset ของกลุ่มตัวเองถึง สภาพ dataset ที่สามารถเป็นไปได้

Lab Overview & Setup

เป้าหมาย: สร้างฟังก์ชัน preprocessing ทีละขั้น แล้วประกอบเป็น pipeline ที่นำไปใช้ในโปรเจกต์ทีมได้จริง

- ครอบคลุม: image representation → thresholding → denoise → convolution/edge → geometric transforms
- ในความเป็นจริง user ไม่ได้อัปภาพหรือวิดีโอที่สวยงามเสมอไปมาให้เรา ดังนั้น เราจะสร้าง dataset ให้สมจริงที่สุด

ติดตั้งเครื่องมือ:

```
pip install opencv-python numpy matplotlib
```

```
import cv2, numpy as np, matplotlib.pyplot as plt
```

Exercise 1 — ภาพคือตัวเลข (Image Representation)

โจทย์: โหลดภาพสี อ่าน shape/จำนวนช่อง แยก R, G, B แล้วแปลง grayscale ด้วยสูตรถ่วงน้ำหนักเอง หรือใช้ฟังก์ชัน เช่น $(0.299R + 0.587G + 0.114B)$ เทียบกับ `cv2.cvtColor`

เฉลย (ย่อ):

```
img = cv2.cvtColor(cv2.imread('doc.jpg'), cv2.COLOR_BGR2RGB)
```

```
R, G, B = img[:, :, 0], img[:, :, 1], img[:, :, 2]
```

```
gray = (0.299*R + 0.587*G + 0.114*B).astype(np.uint8)
```



ภาพ = ฟังก์ชัน $f(x,y)$ คำนวณค่าความเข้ม; ให้น้ำหนักเขียวมากที่สุดเพราะตามมนุษย์ไวต่อสีเขียว

Exercise 2 — Thresholding / Binarization

โจทย์: แยกตัวอักษรดำออกจากพื้นขาวให้คมที่สุด เทียบ 3 วิธี — fixed (127), Otsu (เลือกเกณฑ์อัตโนมัติ), Adaptive (แสงไม่สม่ำเสมอ) ถ้าภาพมีเงาควรใช้อันไหน?

เฉลย (ย่อ):

```
_ , otsu = cv2.threshold(g, 0, 255, cv2.THRESH_BINARY_INV+cv2.THRESH_OTSU)
```

```
adapt = cv2.adaptiveThreshold(g, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
```

```
cv2.THRESH_BINARY_INV, 31, 10)
```



► Adaptive ชนะ (คำนวณเกณฑ์ต่อพื้นที่); fixed/Otsu ใช้เกณฑ์

Exercise 3 — กำจัด Noise (Mean vs Median)

โจทย์: เติม noise แบบ salt-and-pepper เอง แล้วลบด้วย Mean (averaging) เทียบ Median — อันไหนรักษาขอบตัวอักษรได้ดีกว่า และทำไม?

เฉลย (ย่อ):

```
mean_f = cv2.blur(noisy, (3,3))    # low-pass / averaging
```

```
median_f = cv2.medianBlur(noisy, 3) # non-linear
```



— เลือกค่ากลางในหน้าต่าง จุดดำสุดโต่งถูกตัดออก
อักษรเบรลล์

Exercise 4 — Convolution & Edge Detection

โจทย์: เขียน convolution 2D เอง (kernel เฉลี่ย 3×3) เทียบกับ cv2.filter2D แล้วใช้ Sobel หาขอบเพื่อระบุบริเวณตัวอักษร/เส้น (ความถี่สูง)

เฉลย (ย่อ):

```
sx = cv2.Sobel(g, cv2.CV_64F, 1, 0, ksize=3) # ขอบแนวตั้ง
```

```
sy = cv2.Sobel(g, cv2.CV_64F, 0, 1, ksize=3) # ขอบแนวนอน
```

```
edge = cv2.convertScaleAbs(np.hypot(sx, sy))
```



averaging = low-pass (เบลอ), Sobel = high-pass (ดิ่งขอบ); กลไก 'เลื่อน kernel + คูณบวก' คือหัวใจเดียวกับ convolution ใน CNN

Exercise 5 — Deskew เอกสารเอียง (Rotation)

โจทย์: เอกสารสแกนมาเอียง ทำให้ OCR พัง จงหามุมเอียงอัตโนมัติ แล้วหมุนกลับ (rotation matrix) ให้ตรง

เฉลย (ย่อ):

```
M = cv2.getRotationMatrix2D((w//2, h//2), angle, 1.0)
```

```
deskew = cv2.warpAffine(g, M, (w, h),
```

```
flags=cv2.INTER_CUBIC, borderValue=255)
```



cv2.getRotationMatrix2D คือเมทริกซ์หมุน cos/sin ที่พิสูจน์ในคาบ warp เป็น
การ filter ที่เปลี่ยน 'ค่า' (angle)

สิ่งที่ต้องส่ง ใน discord

- ส่งเป็นไฟล์ .py หรือ notebook หนึ่งไฟล์พร้อมภาพ before/after



ในไฟล์จะมีผลลัพธ์ภาพ before/after ทุกชั้น ของแต่ละ exercise เพื่อแสดงว่าเข้าใจว่าแต่ละ filter ทำอะไร

กลุ่ม OCR transcript สามารถเอา dataset ได้จากใน discord

กลุ่ม OCR เล่มหลักสูตรของคณะ IT สามารถเลือกหลักสูตรได้จาก <https://office.kmitl.ac.th/oaq/curriculum/>

